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for the Degree of Bachelor of Science (B.Sc.)

Using Coherent Structures to Reduce Drag Force

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Declaration of Authenticity

I hereby declare that this bachelor thesis titled ”**Using Coherent Structures to Reduce Drag Force**” is my own original work conducted under the supervision of **Dr. Mousavi** at the **Faculty of Mechanical Engineering, Sharif University of Technology**.

All secondary literature, data sources, theoretical formulations, and software tools utilized during this research have been fully cited and acknowledged in accordance with academic integrity and ethical research standards.

Keyhan Javadi

September 2026

*Dedicated to my beloved parents,
whose boundless love, sacrifices, and steadfast encouragement
have been the greatest foundation of my life.*

*And to all the teachers and professors
who ignited and guided my passion for science and engineering.*

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Abstract

Skin-friction drag in wall-bounded turbulent flows is a major source of energy consumption in mechanical and aerospace systems, particularly in internal flows through channels, ducts and pipes and in external boundary layers on aircraft and marine vehicles.[1, 2] The dominant contribution to this drag arises from near-wall coherent structures—quasi-streamwise vortices, low-speed streaks, hairpin packets and larger-scale motions—that generate Reynolds shear stress and transport momentum from the outer flow to the wall.[3, 4] Over the past several decades, a wide range of drag-reduction strategies have been proposed, which can be organized into passive methods (e.g. riblets, superhydrophobic and liquid-infused surfaces, anisotropic porous walls), active methods (e.g. spanwise wall oscillation, travelling waves of spanwise velocity, wall blowing and suction, plasma actuators and feedback control) and hybrid methods that combine multiple mechanisms.[5–15]

This thesis investigates how modification of coherent structures can be exploited systematically for drag reduction, with a particular focus on passive microgroove surfaces in a turbulent rectangular water channel and their relation to the broader literature on coherent-structure-based control. The experimental facility consists of an acrylic channel of width 51.2 mm and height 6.4 mm, corresponding to a hydraulic diameter $D_h \approx 11.4$ mm, with flow of water at 30 °C in the bulk-velocity range $U_b = 0.51–1.19$ m/s, giving turbulent Reynolds numbers $Re_{D_h} \approx 7.2 \times 10^3–1.7 \times 10^4$. Because the channel aspect ratio $w/h \approx 8$ lies outside the recommended range for classical Blasius–Fanning correlations, a Jones-corrected Fanning friction factor is adopted to estimate friction factor, wall shear stress and friction velocity, and viscous sublayer thicknesses $\delta_v = 5\nu/u_\tau$ are computed to guide the design of shark-skin-inspired denticles and microgroove geometries.[16]

This viscous-sublayer analysis shows that while microstructures require strict confinement within the viscous sublayer ($h_d^+ < 5$), the full-height (1.0 x) denticles extended into the buffer and logarithmic layers ($y^+ \approx 15–30$), leading to significant form drag and pressure penalties of 25–35 %. In contrast, truncating denticles to a tip-only profile or employing properly scaled microgrooves confined within the viscous sublayer successfully eliminated bluff-body form drag. Differential pressure measurements acquired at volu-

metric flow rates of 0.6, 1.0 and 1.4 m³/h are used to compare a smooth aluminum base plate, a smooth acrylic reference plate and several microstructured surfaces, including triangular, rectangular, and trapezoidal microgrooves aligned with or perpendicular to the mean flow, as well as shark-denticle textures. Rectangular microgrooves aligned with the flow produce the largest drag reduction, with reductions in pressure drop of approximately 21.8–28.4 % relative to the aluminum base and 21.7–22.7 % relative to the smooth acrylic plate, depending on flow rate, whereas triangular microgrooves in the streamwise direction achieve more modest drag reductions (2.9–15.8 %), and perpendicular grooves as well as full-height denticle configurations significantly increase drag.

Placed in the context of coherent-structure-based drag reduction, the present experiments provide a physically interpretable, manufacturable microgroove geometry that primarily targets near-wall quasi-streamwise vortices and streaks. Direct numerical simulations and experimental work on riblets, superhydrophobic and liquid-infused surfaces, anisotropic porous walls, spanwise wall oscillation and polymer drag reduction have shown that successful drag reduction is closely linked to the suppression or reorganization of near-wall coherent structures, such as the attenuation of ejection events and modification of hairpin packets and large-scale motions.[4, 5, 7, 8, 10, 17] The thesis builds on these insights by (i) reviewing the role of coherent structures in wall-bounded turbulence and their modification by the main classes of drag-reduction techniques, (ii) presenting detailed design and scaling arguments for the microgroove surfaces tested in the rectangular channel, and (iii) discussing how the observed drag reductions can be interpreted in terms of near-wall coherent-structure dynamics and how they benchmark against active and hybrid coherent-structure control strategies.

Keywords: Coherent Structures, Drag Reduction, Wall-Bounded Turbulence, Near-Wall Vortices, Rectangular Water Channel, Microgrooves, Viscous Sublayer, Superhydrophobic Plastron, Cassie–Baxter State.

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List of Symbols

Latin Symbols

Symbol	Description	SI Unit
A	Cross-sectional flow area of the channel	m^2
D_h	Hydraulic diameter of the rectangular channel	m or mm
f_F	Fanning friction factor	–
$f_{F,\text{Jones}}$	Jones-corrected Fanning friction factor	–
h	Channel cross-section height	mm
h_d	Characteristic height/depth of microgrooves or den- ticles	μm or mm
h^+	Dimensionless viscous height of microstructure ($h_d u_\tau / \nu$)	–
L	Effective test section length of the channel	m
p	Static fluid pressure	Pa
Δp	Differential pressure drop across test section	cmH_2O or Pa
Q	Volumetric flow rate of water	m^3/h
Re_{D_h}	Reynolds number based on hydraulic diameter ($U_b D_h / \nu$)	–
Re_τ	Friction Reynolds number ($u_\tau h / 2\nu$)	–
s	Spacing / pitch between microgroove crests	μm or mm
s^+	Dimensionless viscous spacing of microgrooves ($s u_\tau / \nu$)	–
U_b	Bulk mean flow velocity in channel	m/s
u, v, w	Instantaneous velocity components in x, y, z	m/s
u', v', w'	Velocity turbulent fluctuations in x, y, z	m/s
u_τ	Wall friction velocity ($\sqrt{\tau_w / \rho}$)	m/s
w	Channel cross-section width	mm
x, y, z	Streamwise, wall-normal, and spanwise coordinates	m
y^+	Dimensionless wall-normal distance ($y u_\tau / \nu$)	–

Greek Symbols

Symbol	Description	SI Unit
δ_v	Viscous sublayer thickness ($5\nu/u_\tau$)	μm
θ	Static liquid droplet contact angle	Degrees ($^\circ$)
μ	Dynamic viscosity of water	$\text{Pa} \cdot \text{s}$
ν	Kinematic viscosity of water (μ/ρ)	m^2/s
ρ	Density of water	kg/m^3
τ_w	Mean wall shear stress	Pa
$-\rho \overline{u'v'}$	Reynolds shear stress in the channel symmetry plane	Pa

Subscripts and Superscripts

Notation	Definition
$(\cdot)_b$	Bulk mean or cross-sectional average
$(\cdot)_d$	Related to denticle or microgroove elements
$(\cdot)_h$	Based on hydraulic diameter
$(\cdot)_w$	Wall value
$(\cdot)_\tau$	Scaled by friction velocity and wall shear stress
$(\cdot)^+$	Non-dimensionalized in viscous inner wall units

Acronyms and Abbreviations

Acronym	Definition
CFD	Computational Fluid Dynamics
DNS	Direct Numerical Simulation
DR	Drag Reduction Percentage (%)
IPA	Isopropyl Alcohol
LES	Large Eddy Simulation
LIS	Liquid-Infused Surfaces
PIV	Particle Image Velocimetry
Q2	Ejection event ($u' < 0, v' > 0$)
Q4	Sweep event ($u' > 0, v' < 0$)
SHS	Superhydrophobic Surfaces
SLA	Stereolithography (3D Printing)
UV	Ultraviolet Light (Post-Curing)

Chapter 1

Introduction

1.1 Motivation and Engineering Context

Wall-bounded turbulent flows are ubiquitous in mechanical and aerospace engineering, appearing in internal flows through pipes, channels and ducts as well as in external boundary layers over aircraft, turbine blades and marine hulls. Skin-friction drag generated at the fluid–solid interface is often a dominant contribution to the total drag, directly impacting pumping power, fuel consumption and operating costs. Comprehensive reviews have shown that even modest reductions in friction drag can translate into substantial energy savings over the lifetime of industrial transport and process systems.[1, 2]

In internal flows, such as those in heat exchangers, microchannel devices and industrial piping, pumping power scales approximately with the product of volumetric flow rate and pressure drop. For a fixed mass flow rate, any reduction in wall shear stress leads to a lower pressure gradient and reduced power consumption. Conversely, for a fixed pressure gradient, drag reduction enables an increased flow rate, which may improve heat-transfer performance or throughput. Similar trade-offs apply in external flows: reducing skin-friction drag on aircraft and marine vessels improves propulsive efficiency and can provide environmental benefits through lower emissions.

The growing focus on energy efficiency and sustainability has therefore motivated extensive research into methods for controlling turbulent flows and reducing skin-friction drag. These methods range from purely passive surface treatments to sophisticated active control strategies and hybrid combinations. Crucially, the success of many drag-reduction techniques is closely linked to how they modify the coherent structures that populate wall-bounded turbulence. Understanding these structures and their response to control is central to the present thesis.

1.2 Wall-Bounded Turbulence and Coherent Structures

Turbulence in wall-bounded flows exhibits strongly organized motions near the wall, often referred to as coherent structures. Even though the flow is chaotic and multi-scale, certain features—including quasi-streamwise vortices, low-speed streaks, hairpin vortices and larger-scale motions—appear repeatedly and play a disproportionate role in producing Reynolds shear stress and skin-friction drag.[3, 4]

In canonical turbulent channel and pipe flows, the near-wall region is traditionally divided into the viscous sublayer, buffer layer and logarithmic region based on the wall-normal coordinate expressed in viscous units. Near the wall, in the viscous sublayer, the flow is approximately laminar, and viscous effects dominate. Further away, in the buffer layer, quasi-streamwise vortices and streaks drive intense turbulent activity. These structures extract energy from the mean shear and redistribute momentum in the wall-normal direction. The logarithmic region, extending further from the wall, is populated by larger-scale and very-large-scale motions that interact with the near-wall cycle and modulate its dynamics.[3]

Quasi-streamwise vortices are elongated structures aligned roughly with the mean flow direction, inducing strong wall-normal and spanwise velocity fluctuations. Their rotation generates regions of low-speed fluid lifted away from the wall (ejections) and high-speed fluid pushed towards the wall (sweeps). Low-speed streaks are long, narrow regions of reduced streamwise velocity that form between vortices and are often sites of instability and bursting. Hairpin vortices and packets—arching structures whose legs are attached near the wall—are associated with organized turbulent events, including the ejection of low-speed fluid away from the wall. These coherent structures collectively sustain the turbulence regeneration cycle and drive Reynolds shear stress and friction drag.[4]

From the perspective of drag reduction, coherent structures are both the problem and the solution. On the one hand, their dynamics produce the turbulent mixing that increases skin-friction drag relative to laminar flow. On the other hand, many control strategies—passive, active and hybrid—work precisely because they alter the formation, strength, spatial organization or temporal behavior of these structures. For example, riblets can inhibit crossflow and weaken quasi-streamwise vortices, polymer additives can damp turbulent bursts and hibernation intervals, and spanwise wall oscillations can disrupt the near-wall cycle by inducing an oscillatory shear layer. This thesis adopts coherent structures as the central lens through which drag reduction is analyzed.

1.3 Classification of Drag-Reduction Strategies

Drag-reduction methods in wall-bounded turbulence can be broadly classified into three categories: passive, active and hybrid strategies.[1, 2]

1.3.1 Passive Methods

Passive methods rely on fixed modifications to the wall or fluid that do not require continuous external actuation once implemented. Classic examples include:

- **Riblets and microgrooves**, which are small streamwise grooves designed to interfere with crossflow and reduce turbulent momentum transfer within the viscous sublayer.[2, 5]
- **Superhydrophobic surfaces**, which trap gas in surface textures and create effective slip at the wall, reducing shear stress and pressure drop.[6, 7, 15]
- **Liquid-infused surfaces**, where an immiscible liquid fills surface grooves to provide a lubricated interface and effective slip.
- **Anisotropic porous walls**, which provide streamwise-preferential permeability and slip while limiting wall-normal penetration of turbulence.[8, 9]
- **Roughness and bio-inspired surfaces**, including shark-skin-inspired textures that can reduce drag by organizing near-wall coherent structures.[5]
- **Additive-based methods**, such as polymer and surfactant drag reduction, where small amounts of additives modify the fluid rheology and suppress turbulent bursts.[3]

These methods are attractive for technological applications because they do not require sensors, actuators or control algorithms once installed. However, their performance depends critically on geometric parameters (height, spacing, shape), surface chemistry, contamination and operational Reynolds number.

1.3.2 Active Methods

Active drag-reduction strategies use actuators to directly manipulate the flow in real time or open loop. Notable examples include:

- **Spanwise wall oscillation**, in which the wall or surface elements oscillate in the spanwise direction, generating an oscillatory Stokes layer that modifies near-wall coherent structures and can produce large drag reductions.[10, 11, 17]
- **Streamwise-travelling waves of spanwise velocity**, where a travelling wave of transverse velocity is imposed on the wall to disrupt the turbulence regeneration cycle.[10]
- **Wall blowing and suction**, including zero-net-mass-flux jets and travelling waves of blowing/suction, which alter mean velocity and turbulent structures near the wall.[12]
- **Plasma actuators**, particularly dielectric-barrier-discharge devices that produce transverse forcing near the wall without moving parts.
- **Feedback and learning-based control**, where sensors measure flow quantities and actuators respond according to control laws designed via optimal control or deep reinforcement learning.[12, 13]

These methods can achieve very large drag reductions—often exceeding 40–50% in idealized configurations—but their practical value depends on actuator power consumption, reliability, integration and control complexity.

1.3.3 Hybrid Methods

Hybrid methods combine passive and active features or multiple passive elements. Examples include:

- **Polymer additives with superhydrophobic surfaces**, where the fluid rheology and wall slip are simultaneously modified.[14, 15]
- **Riblet–superhydrophobic composites**, in which riblet geometry organizes near-wall structures while superhydrophobic coatings provide slip.
- **Porous walls with spanwise forcing**, combining permeability and active transverse motion.

Hybrid strategies can either enhance or compromise drag reduction, depending on how the underlying mechanisms interact. They highlight the importance of a unifying perspective—such as coherent structures—for understanding and designing drag-reduction techniques.

1.4 Drag Reduction through the Lens of Coherent Structures

A key insight of recent research is that successful drag-reduction methods develop when considered through the lens of coherent structures rather than purely in terms of mean profiles or global coefficients. This section outlines how several major techniques affect coherent structures and thereby reduce drag.

1.4.1 Riblets and Microgrooves

Riblets and microgrooves are streamwise-oriented surface features designed to interact with near-wall coherent structures in the viscous sublayer and buffer layer. Experiments by Bechert and co-workers and subsequent DNS studies showed that properly sized riblets can reduce skin-friction drag by about 5–10%, with geometry and alignment playing crucial roles.[\[2, 5\]](#)

The primary mechanism is the suppression of crossflow and interference with quasi-streamwise vortices. When riblet height and spacing are confined within the viscous sublayer, the grooves restrict spanwise motion and reduce the ability of vortices to transport momentum towards the wall. The groove tips act as barriers that limit lateral displacement of low-speed streaks and damp turbulent ejection events. If riblets become too large or are misaligned, however, they can generate secondary flows and spanwise-coherent rollers, increasing drag.

Microgrooves—such as the triangular and rectangular grooves in this thesis—can be viewed as riblet variants with cross-sections adapted to manufacturing constraints and channel geometry. By designing microgroove height based on viscous-sublayer thickness, microgrooves can be tuned to interact with the near-wall cycle in a beneficial way. The present experiments aim to quantify their drag-reduction performance and interpret the results using coherent-structure concepts.

1.4.2 Superhydrophobic and Liquid-Infused Surfaces

Superhydrophobic surfaces (SHSs) and liquid-infused surfaces (LISs) reduce drag by introducing effective slip at the wall. Fukagata, Kasagi and Koumoutsakos developed a theoretical framework for friction drag reduction in turbulent flow over superhydrophobic walls, relating drag reduction to slip length and gas fraction.[\[6\]](#) Rothstein and others experimentally confirmed that streamwise grooves filled with trapped gas can significantly

reduce wall shear stress, although surfactant contamination and loss of the gas layer can negate the effect.[7, 15]

From a coherent-structure standpoint, SHSs and LISs modify the near-wall condition such that quasi-streamwise vortices interact with a partially slipping interface rather than a strict no-slip wall. This reduces the production of near-wall turbulent kinetic energy and alters the balance of ejections and sweeps. At the same time, the presence of a multi-phase interface introduces additional phenomena (e.g. capillary waves and interface deformation) that can create new structures and affect drag.

1.4.3 Anisotropic Porous Walls

Anisotropic porous walls, studied by Rosti, Brandt, Pinelli and others, provide streamwise-preferential slip while limiting wall-normal flow through tailored permeability tensors.[8, 9] These substrates can reduce drag when streamwise permeability is large and wall-normal permeability is small, suppressing wall-normal fluctuations and promoting tangential slip. However, if wall-normal permeability is too large, Kelvin–Helmholtz-like rollers can form, increasing drag.

The coherent-structure interpretation is that porous substrates reshape the near-wall turbulence regeneration cycle by changing how quasi-streamwise vortices and streaks exchange fluid with the wall. Drag reduction corresponds to a regime in which the near-wall cycle is weakened and structures become more “one-component,” while drag increase arises when new spanwise-coherent structures dominate.

1.4.4 Spanwise Wall Oscillation and Travelling Waves

Spanwise wall oscillation and streamwise-travelling waves of spanwise velocity are among the most effective active drag-reduction techniques. Quadrio, Ricco and Viotti showed that streamwise-travelling waves of spanwise wall velocity can significantly reduce skin-friction drag in turbulent channel flow.[10] Gatti and Quadrio quantified the Reynolds-number dependence of such control, reporting drag-reduction levels that remain promising even at high Reynolds numbers.[11] Marusic and co-workers demonstrated an energy-efficient pathway to drag reduction by targeting large-scale motions in high-Reynolds-number boundary layers.[17]

These methods work by imposing an oscillatory or travelling shear layer at the wall, which alters the formation, orientation and lifetime of near-wall coherent structures. Quasi-streamwise vortices and streaks are subjected to a time-dependent transverse forcing that

disrupts their ability to regenerate and sustain turbulence. DNS and conditional averaging studies have shown that spanwise forcing weakens ejection events and modifies the phase relationships between inner and outer scales, leading to reduced Reynolds shear stress.

1.4.5 Polymer Drag Reduction and Coherent Structures

Polymer drag reduction (PDR) is another technique whose success is closely tied to coherent structures. Xi reviewed the fundamentals and recent advances in turbulent drag reduction by polymer additives, while Saeed, Xi and White examined PDR through the lens of coherent structures.[3, 4] Polymer molecules, when sufficiently long and flexible, stretch in response to turbulent fluctuations and exert elastic stresses that oppose velocity gradients and suppress turbulent bursts.

Coherent-structure analysis shows that polymers damp the near-wall regeneration cycle by attenuating Q2 ejections and reducing the intensity of hairpin packets and streak instabilities. At high drag-reduction levels, the turbulence can enter “hibernation” intervals characterized by weak coherent structures and near-laminar behavior. Understanding these dynamics is crucial for designing PDR systems and for drawing analogies with other coherent-structure-based control methods.

1.5 Thesis Focus: Passive Microgrooves and Coherent Structures

Against this broad background, the present thesis focuses on passive microgroove surfaces in a turbulent rectangular water channel and interprets their drag-reduction performance through coherent structures.

The experimental channel has a width of 51.2 mm and a height of 6.4 mm, yielding a hydraulic diameter of approximately 11.4 mm. Water at 30 °C flows through the channel at bulk velocities in the range 0.1–1.18 m/s, corresponding to turbulent Reynolds numbers. Because the channel aspect ratio lies outside the regime where classical Blasius–Fanning correlations are most accurate, a Jones-corrected Fanning friction factor correlation is employed to estimate wall shear stress and friction velocity. The viscous-sublayer thickness is then computed to guide microgroove height selection, ensuring that groove dimensions remain within the viscous sublayer where they are most effective.

Microgroove surfaces are fabricated using high-resolution 3D printing based on shark-skin-inspired denticle geometries and simplified triangular and rectangular groove patterns.

Manufacturing constraints—such as minimum printable feature height on the Elegoo Saturn 4 Ultra 16 K system—are explicitly considered, leading to the choice of an original 1.0x scale for denticle height and the exclusion of both larger scales, which would protrude beyond the viscous sublayer, and smaller scales, which would be below the printer’s resolution limit.

Differential pressure measurements at multiple flow rates are used to quantify drag reduction relative to smooth aluminum and acrylic baseline plates. Preliminary results show that rectangular microgrooves aligned with the flow yield the largest drag reduction, while triangular grooves and perpendicular configurations produce more modest reductions or even drag increases. These observations align qualitatively with riblet theory and coherent-structure-based interpretations: streamwise-aligned, shallow grooves suppress crossflow and weaken quasi-streamwise vortices, whereas misaligned or overscaled grooves can generate new coherent structures that increase drag.[\[2, 5\]](#)

1.6 Structure and Objectives of the Thesis

The overarching objective of this thesis is to use coherent structures as a unifying framework for understanding drag reduction and to apply this framework to the design, interpretation and optimization of passive microgroove surfaces in a turbulent rectangular channel.

More specifically, the thesis aims to:

1. Review the physics of wall-bounded turbulence and the role of coherent structures in generating skin-friction drag, emphasizing quasi-streamwise vortices, streaks, hairpins and large-scale motions.
2. Survey major drag-reduction strategies—passive, active and hybrid—with particular attention to how each method modifies coherent structures, drawing on riblet, superhydrophobic, porous-wall, spanwise forcing and polymer drag-reduction literature.
3. Present the design and scaling of shark-skin-inspired denticles and microgroove surfaces for the rectangular water channel, including hydraulic calculations, viscous-sublayer analysis and manufacturing constraints.
4. Report and analyze differential-pressure measurements for various surface configurations, quantifying drag reduction and relating the results to coherent-structure behavior.

5. Discuss how the insights gained from passive microgrooves compare with those from other coherent-structure-based control techniques and outline directions for future work, including potential hybrid strategies and active control in similar geometries.

The remainder of the thesis is organized as follows. Chapter 2 (Related and Past Works) provides a detailed literature review on coherent structures and drag reduction across the main classes of control techniques. Chapter 3 describes the experimental facility, microgroove design and measurement techniques. Chapter 4 presents the experimental results and their interpretation in terms of coherent structures. Finally, Chapter 5 summarizes the findings and discusses implications for the design of coherent-structure-based drag-reduction strategies in mechanical engineering applications. The chapter below is based on published studies on riblets and near-wall coherent structures, spanwise wall oscillation, and polymer drag reduction. ([Springer Link](#))

Chapter 2

Related and Past Works

2.1 Introduction to Coherent-Structure-Based Drag Reduction

Drag reduction in wall-bounded turbulent flows has been a central topic in fluid mechanics for over seventy years, driven by its importance for energy-efficient transport and industrial processes.[1, 3] Classical work established empirical correlations and control concepts, but more recent research has increasingly focused on coherent structures—organized turbulent motions such as quasi-streamwise vortices, low-speed streaks, hairpins and large-scale motions—and on how manipulating these structures can lead to reduced skin-friction drag.[4]

This chapter reviews past and related works on drag reduction with an emphasis on coherent-structure mechanisms. The goal is not only to summarize results, but to highlight how different methods alter the near-wall regeneration cycle, redistribute Reynolds stresses and change the energy pathways in wall-bounded turbulence. The discussion is organized into several major themes:

- Coherent structures in wall-bounded turbulence and their role in drag production.
- Passive control strategies: riblets and microgrooves, superhydrophobic and liquid-infused surfaces, anisotropic porous walls, and polymer/surfactant additives.
- Active control strategies: spanwise wall oscillation, streamwise-travelling waves, wall blowing and suction, and feedback or learning-based control.
- Hybrid strategies that combine multiple mechanisms, especially polymers with textured walls.

The chapter concludes by synthesizing these mechanisms into a coherent-structure-based framework that underpins the passive microgroove study in this thesis.

2.2 Coherent Structures in Wall-Bounded Turbulence

2.2.1 Near-Wall Regeneration Cycle

In fully developed turbulent channel and pipe flows, the dominant contribution to skin-friction drag comes from coherent structures in the near-wall region. The buffer layer (roughly $30 \lesssim y^+ \lesssim 100$) and adjacent viscous sublayer host a self-sustaining regeneration cycle consisting of quasi-streamwise vortices, low-speed streaks, and intermittent bursting events.[3, 4]

Quasi-streamwise vortices are elongated structures aligned approximately with the mean flow direction. They induce strong wall-normal and spanwise velocity fluctuations, giving rise to the characteristic Q2 (ejection) and Q4 (sweep) events in quadrant analysis of Reynolds shear stress. Low-speed streaks are long, narrow regions of reduced streamwise velocity that form between vortices; they can extend for hundreds of viscous units in the streamwise direction. Hairpin vortices, whose legs are attached near the wall and whose arches extend outward, often form organized “packets” that bridge the buffer and logarithmic regions.

The regeneration cycle can be summarized as follows:

1. Streamwise vortices lift low-speed fluid away from the wall, creating low-speed streaks and Q2 ejections.
2. Streaks become unstable, spawning new vortices and hairpins.
3. These structures extract energy from the mean shear, generate Reynolds shear stress and transport momentum towards the wall.
4. The decay of structures completes the cycle, which then repeats.

Any drag-reduction technique that significantly alters this cycle—for example, by weakening the vortices, changing streak spacing, or modifying the timing and strength of ejections and sweeps—can reduce skin-friction drag.

2.2.2 Large-Scale and Very-Large-Scale Motions

Beyond the buffer layer, the logarithmic and outer regions host large-scale and very-large-scale motions (LSMs and VLSMs). These structures, which can extend over many boundary-layer thicknesses, interact with the near-wall cycle via amplitude and phase modulation.[3] For instance, high-speed LSMs can increase the frequency or intensity of near-wall bursts, while low-speed LSMs can suppress them.

Recent work on drag reduction has emphasized that control strategies targeting large-scale motions can be more energy-efficient than those directly forcing small-scale structures. Marusic and co-workers showed that low-frequency spanwise forcing tuned to large-scale motions in high-Reynolds-number boundary layers can produce substantial drag reductions while minimizing actuation power.[17] This suggests that coherent structures across scales—and their interactions—must be considered when designing control methods.

2.2.3 Coherent Structures as a Design Lens

Saeed, Xi and White argued that polymer drag reduction is best understood by examining how polymer chains alter coherent structures, rather than only looking at mean profiles or global friction factors.[4] They developed a polymer-modified version of the near-wall cycle, showing how chains stretch in streaks, relax in vortices and exert elastic stresses that weaken the self-sustaining process.

This coherent-structure lens has since been extended to other drag-reduction methods. Riblets, superhydrophobic surfaces, porous walls and spanwise forcing have all been studied in terms of how they reshape vortical structures, streak patterns and the distribution of Q2/Q4 events.[5–8, 10, 11] The present thesis adopts this perspective as a unifying framework for passive microgroove design and analysis.

2.3 Passive Control: Riblets and Microgrooves

2.3.1 Classical Riblet Experiments

Riblets are streamwise-invariant grooves and ridges designed to reduce turbulent skin-friction drag by interfering with crossflow near the wall. Bechert et al. carried out landmark experiments in an oil channel to systematically optimize riblet geometries.[5] Using adjustable triangular, semi-circular and trapezoidal grooves, they mapped drag reduction as a function of riblet height and spacing in viscous units.

Key findings from their work include:

- Optimal riblets produced skin-friction reductions of about 5–10%, with a record value near 9.9% for trapezoidal grooves.
- Drag reduction was achieved when riblet height was confined within the viscous sublayer and lateral spacing matched characteristic streak spacing.
- Oversized riblets or poorly aligned grooves caused drag increase due to secondary flows and increased pressure drag.

Their experiments supported Luchini’s theoretical model, in which riblets are assumed to impede the fluctuating crossflow near the wall, reducing momentum transfer and shear stress.^[5]

2.3.2 DNS and Coherent Structure Analysis over Riblets

Direct numerical simulations (DNS) have provided detailed insight into how riblets affect coherent structures. Recent minimal-channel DNS studies considered a variety of riblet geometries (triangular, blade, trapezoidal) and sizes, quantifying their impact on near-wall turbulence.^[2] Dispersive stresses—spatial averages of fluctuations induced by the surface—were used to interpret drag changes, revealing that riblets alter the spatial distribution and coherence of vortical structures.

Two-point statistics and three-dimensional spatial correlations have been used to analyze coherent structures over riblets. Huang et al. performed DNS of turbulent flow over both drag-reducing and drag-increasing riblet configurations and studied the effects on near-wall coherent structures using correlation analysis.^[18] Their results showed that:

- Drag-reducing riblets shortened the streamwise extent of near-wall vortices and weakened their intensity.
- Drag-increasing riblets amplified certain vortical motions and generated spanwise-coherent structures resembling Kelvin–Helmholtz rollers.
- Changes in coherent structures were directly linked to changes in Reynolds shear stress and drag.

Riblet geometry can thus be seen as a way to “filter” coherent structures: certain shapes suppress harmful motions, while others inadvertently promote them.

2.3.3 Riblet Drag-Reduction Models

Because DNS of riblets at engineering Reynolds numbers is expensive, several reduced models have been proposed. A viscous vortex model introduced by recent work extracts the footprint of a typical quasi-streamwise vortex from smooth-wall DNS fields and solves a Stokes-flow problem with embedded riblets.[19] By changing riblet cross-section and size, the model predicts how vortices interact with the grooves and estimates drag reduction without full DNS.

Such models highlight that drag-reducing riblets operate by modifying vortex footprints. If riblet tips lie at locations where vortices would otherwise generate strong crossflow into the wall, they reduce vortex-induced shear. Conversely, riblets that intersect coherent structures unfavorably can generate new shear layers and increase drag.

2.3.4 Microgrooves and Flow-Field Studies

Beyond idealized riblets, practical surfaces often employ microgroove patterns adapted to manufacturing and structural constraints. Numerical studies of microgrooves in turbulent channels and boundary layers have examined how these grooves alter flow fields, vorticity and coherent structures. Scalloped riblets, for example, have been shown to reduce drag more than corresponding triangular riblets, with drag-reduction rates around 5.77% at optimal groove width and height-to-width ratio.[20]

Flow-field analysis using second-order statistics of velocity, vorticity and vortex-identification metrics (e.g. Liutex) has revealed that:

- Streamwise vortices just above riblet tips are shorter and weaker, with length scales reduced to 200–300 wall units in some designs.
- The spatial distribution of coherent structures is reorganized, such that high-shear events are less frequent or less intense near the wall.
- Drag-reducing riblets tend to reduce the probability and strength of Q2 ejections, while drag-increasing riblets can enhance Q4 sweeps or generate large-scale rollers.[18, 20]

These studies demonstrate how microgroove geometry, even when deviating from classical riblets, can be tuned to target specific coherent structures.

2.3.5 Riblets in Adverse-Pressure-Gradient Flows

Recent DNS work has investigated riblets in separating boundary layers subject to adverse pressure gradients (APG). In such flows, riblets that increase drag under zero-pressure-gradient conditions have been found to reduce drag under APG by generating Kelvin–Helmholtz rollers that shorten separation bubbles and relieve pressure drag.[21] Riblets can thus influence both friction and pressure drag components via coherent-structure modulation.

Application of riblets to separating turbulent boundary layers shows that:

- Riblet-generated KH rollers can create patches of flow reversal that change mean separation distance.
- Drag reduction can reach very high values (up to 100% in some configurations) when riblets are properly tuned to APG conditions.[21]

This underscores the importance of considering coherent structures beyond the near-wall cycle, especially in flows where separation and pressure drag dominate.

2.4 Passive Control: Superhydrophobic and Liquid-Infused Surfaces

2.4.1 Superhydrophobic Surfaces and Slip

Superhydrophobic surfaces (SHSs) use micro- or nano-textures to trap gas between surface features, creating an approximately shear-free interface. Fukagata, Kasagi and Koumoutsakos developed a theoretical prediction for drag reduction over superhydrophobic surfaces in turbulent channel flow, expressing drag reduction in terms of mean slip length and gas fraction.[6] They showed that even modest slip lengths can yield measurable reductions in friction factor.

Rothstein reviewed experimental and numerical work on SHSs, highlighting that:

- Streamwise grooves filled with trapped gas can reduce drag by decreasing effective wetted area and introducing slip.
- Surface patterning, gas fraction and interface cleanliness are critical to performance.

- Surfactants or contaminants at the gas–liquid interface can dramatically reduce slip and negate drag reduction.[7]

From a coherent-structure perspective, SHSs modify the near-wall boundary condition so that quasi-streamwise vortices and streaks encounter a partially slipping interface. This reduces the production of turbulent kinetic energy in the viscous sublayer and buffer layer.

2.4.2 Coherent Structure Effects over SHSs

Coherent structure analysis over SHSs has shown that:

- Near-wall vortices are weaker and their rotation is altered by slip at the interface.
- Low-speed streaks are less tightly anchored to the wall, potentially changing their spacing and instability characteristics.
- Q2 ejection events are less intense, contributing to reduced Reynolds shear stress.[15]

However, the presence of a multi-phase interface also introduces new phenomena. Capillary waves and interface deformation can create additional structures that interact with turbulence. Successful SHS designs must ensure that such phenomena do not dominate and that slip-induced drag reduction outweighs any adverse effects.

2.4.3 Liquid-Infused Surfaces

Liquid-infused surfaces (LISs) replace trapped gas with an immiscible lubricant liquid in the grooves. This design often provides better pressure stability than gas, especially at high speeds or in large-scale applications. Experiments have demonstrated drag reduction exceeding 35% by optimizing lubricant viscosity, groove width and fluid area fraction.[15]

Coherent structures in LIS flows interact with a liquid–liquid interface that offers slip but has different inertia and capillarity than a gas–liquid interface. The effect on near-wall vortices and streaks is similar in qualitative terms—reduced shear and altered regeneration cycles—but details differ due to interface properties.

LIS studies show that:

- Groove-scale recirculation within the lubricant can enhance heat transfer while maintaining drag reduction.
- Interface stability (against shear-driven drainage and roughening) is a key design constraint.

2.5 Passive Control: Anisotropic Porous Walls

2.5.1 Drag Increase and Reduction via Permeability Tensors

Anisotropic porous walls offer another passive route to drag modification. Rosti, Brandt and Pinelli studied turbulent channel flows over anisotropic porous walls, varying the permeability tensor to achieve either drag reduction or drag increase.[8] They found that:

- High streamwise permeability with low wall-normal permeability promotes tangential slip and can reduce drag by more than 20%.
- High wall-normal permeability enhances wall-normal fluctuations and can lead to strong drag increase due to spanwise-coherent rollers.[8]

Gomez-de-Segura and García-Mayoral explored similar substrates and developed analytical and resolvent-based models to predict drag trends.[9] Their work showed that drag reduction is proportional to the difference between streamwise and spanwise permeabilities up to a critical wall-normal permeability, beyond which Kelvin–Helmholtz-like instabilities degrade performance.

2.5.2 Coherent Structures over Porous Substrates

Porous substrates change coherent structures by allowing fluid to penetrate the wall-normal direction. When wall-normal permeability is low, near-wall vortices and streaks experience limited penetration and increased tangential slip, weakening the regeneration cycle. When wall-normal permeability is high, new coherent structures resembling KH rollers form, often spanning large wall-parallel extents.

Coherent-structure analysis over porous walls reveals that:

- Drag-reducing substrates produce more “one-component” turbulence near the wall, with reduced wall-normal fluctuations.
- Drag-increasing substrates support spanwise-coherent rollers that dominate the near-wall region and significantly increase Reynolds shear stress.[8, 9]

These observations illustrate the delicate balance between slip and permeability in coherent-structure-based design.

2.6 Passive Control: Polymer and Surfactant Additives

2.6.1 Fundamentals of Polymer Drag Reduction

Since the 1940s, it has been known that adding small amounts of long-chain polymers to turbulent flows can reduce friction drag by up to 80%.[\[3, 22\]](#) Xi reviewed fundamentals and recent advances in polymer drag reduction, emphasizing that polymers stretch in response to velocity gradients and exert elastic stresses that oppose turbulent motions.[\[3\]](#)

Key ideas include:

- Polymers must have sufficient chain extensibility and relaxation times (high Weissenberg numbers) to interact effectively with turbulent structures.
- Drag reduction is often observed first in the buffer layer, where polymers suppress near-wall bursts.
- At high drag-reduction levels, flows can approach the maximum drag reduction (MDR) asymptote, with profoundly modified turbulence structure.

2.6.2 Coherent Structures and Polymer Dynamics

Saeed, Xi and White examined polymer drag reduction through the lens of coherent structures.[\[4\]](#) They combined literature on Newtonian coherent motions with observations of polymer chain dynamics to construct a polymer-modified near-wall cycle.

Their analysis shows that:

- Polymer chains stretch predominantly in streamwise streaks where flow is elongational.
- As polymers move into vortical regions, they relax and exert forces that oppose vortex rotation.
- This relaxation weakens counter-rotating vortex pairs and reduces the intensity of Q2 ejections and Q4 sweeps.

Stone et al. studied polymer drag reduction in “exact coherent structures”—travelling-wave solutions of the Navier–Stokes equations that resemble near-wall turbulent structures.[\[23\]](#)

They found that polymers alter these structures by suppressing streamwise vortices, increasing streamwise velocity fluctuations and decreasing wall-normal fluctuations. The mechanism is consistent with the idea that polymer relaxation opposes vortex motion, leading to reduced drag.

2.6.3 DNS and RANS Studies of Polymer Flows

DNS studies of polymer-laden turbulent boundary layers and channel flows have used constitutive models like FENE-P and Oldroyd-B to represent polymer dynamics.[3, 22] They have shown that:

- Significant drag reduction requires a combination of chain extensibility and high Weissenberg numbers.
- Near-wall turbulence becomes intermittent, with hibernation intervals where coherent structures are weak and drag is reduced.
- Energy budgets reveal a transfer from turbulent kinetic energy to elastic energy stored in polymers.

RANS models have been developed to approximate viscoelastic effects with reduced computational cost. While they often overpredict drag reduction due to smoothing out coherent structures, they provide insight into parameter dependencies and can guide engineering design.[22]

2.6.4 Surfactant Additives

Surfactants can also reduce drag by forming micelles and modifying fluid rheology. The mechanisms are less well understood than for polymers, but evidence suggests that surfactants can alter coherent structures by changing effective viscosity and promoting laminarization in parts of the flow.[1] Surfactants also complicate SHS performance by accumulating at interfaces, reducing slip and altering coherent-structure interactions with the wall.[7]

2.7 Active Control: Spanwise Forcing and Beyond

2.7.1 Spanwise Wall Oscillation

Jung, Mangiavacchi and Akhavan demonstrated that high-frequency spanwise wall oscillation can suppress turbulence and reduce drag in wall-bounded flows.[24] By imposing sinusoidal motion in the spanwise direction, they created an oscillatory Stokes layer that interacts with near-wall structures.

Later work applied conditional averaging techniques to quasi-streamwise vortices in spanwise-oscillated channel flows.[25] Using the Fukagata–Iwamoto–Kasagi (FIK) identity to decompose skin friction into Reynolds-stress contributions, they found that:

- Suppression of Q2 ejection events dominates drag reduction at relatively small oscillation periods.
- Q4 sweep events can contribute to drag reduction at small periods but tend to cause drag increase at larger periods.
- Spanwise oscillation tilts coherent structures and modifies their phase relative to wall motion, controlling the net effect on drag.[25]

These results show that spanwise wall oscillation works by selectively suppressing certain coherent events while potentially enhancing others.

2.7.2 Streamwise-Travelling Waves of Spanwise Velocity

Quadrio, Ricco and Viotti introduced streamwise-travelling waves of spanwise wall velocity (StTW) as a generalization of pure oscillation.[10] DNS studies demonstrated substantial drag reductions, and subsequent analysis by Gallorini and co-workers examined the effect of StTW on quasi-streamwise vortices.[26]

Conditional averaging around vortices showed that:

- StTW modify the quadrant contributions to Reynolds shear stress in ways similar to oscillation but with additional spatial structure.
- Suppression of Q2 events still governs drag reduction, but the phase and spatial distribution of suppression depend on wave parameters.

- Coherent vortices experience alternating shear that tilts and dissipates them, weakening the near-wall cycle.[26]

These studies reinforce the view that spanwise forcing is fundamentally a coherent-structure control strategy.

2.7.3 Energy-Efficient Forcing and Large-Scale Motions

Gatti and Quadrio analyzed how drag-reduction performance of spanwise forcing depends on Reynolds number.[11] They found that drag reduction tends to decrease with increasing Reynolds number, but remains significant over a wide range. They also showed that the effect of forcing can be represented as a vertical shift of the logarithmic mean velocity profile, linking it to classic wall-function theory.

Marusic et al. proposed an energy-efficient pathway to drag reduction by targeting large-scale motions in the outer region rather than directly forcing near-wall structures.[17] By tuning forcing frequency and amplitude to large-scale structures, they were able to maintain substantial drag reduction at high friction Reynolds numbers (Re_τ up to 12 800) with reduced actuation power. This approach leverages coherent-structure interactions across scales.

2.7.4 Wall Blowing and Suction; Plasma Actuators

Choi, Moin and Kim surveyed active turbulence control methods, including wall blowing and suction and plasma actuators.[12] Blowing and suction alter coherent structures by injecting or removing fluid, changing streak spacing and vortex strength. Plasma actuators—especially dielectric-barrier-discharge devices—produce localized body forces that can mimic spanwise or streamwise forcing without moving solid surfaces.

From a coherent-structure standpoint:

- Blowing tends to thicken the local boundary layer, affecting streak formation and vortex location.
- Suction thins the layer and can intensify near-wall turbulence unless carefully controlled.
- Plasma forcing can generate transverse motions that disrupt coherent structures in ways similar to mechanical forcing.[12]

2.8 Active and Learning-Based Control

2.8.1 Classical Feedback Control

Feedback control strategies use sensors and actuators to dynamically adjust the flow based on measured quantities such as wall shear, pressure or velocity. Opposition control, for example, aims to cancel wall-normal velocity fluctuations by applying blowing or suction in opposition to measured disturbances.^[12]

While such methods can be effective in low-Reynolds-number simulations, practical implementation faces challenges: sensor placement, actuator bandwidth and robustness to disturbances. Nevertheless, they illustrate the potential of closed-loop coherent-structure control.

2.8.2 Deep Reinforcement Learning for Turbulent Channels

Guastoni et al. introduced a deep reinforcement learning (DRL) framework for turbulent drag reduction in channel flows.^[13] An agent interacts with the flow via wall-normal blowing/suction actuators and receives rewards based on drag reduction. Over training episodes, the agent learns control laws that modify coherent structures to reduce drag.

DRL-based control offers several advantages:

- It can discover non-intuitive strategies that exploit complex coherent-structure dynamics.
- It can adapt to changing flow conditions or constraints.
- It provides a way to integrate data-driven and physics-based approaches.^[13]

Coherent-structure analysis of DRL-controlled flows shows changes in vortex strength, streak patterns and Reynolds stress distributions, confirming that learned policies operate by restructuring turbulence.

2.9 Hybrid Control: Combining Mechanisms

2.9.1 Polymer Additives with Superhydrophobic Surfaces

Hybrid control strategies combine multiple mechanisms to potentially achieve greater drag reduction than any single method. Rajappan and McKinley studied cooperative

drag reduction in turbulent flows using polymer additives and superhydrophobic walls.[14] They found that polymer additives and SHSs can interact in complex ways:

- In some cases, drag-reduction effects are additive, with polymers and SHSs contributing independently.
- In other cases, one mechanism dominates, and the marginal benefit of the other is small.
- Under certain conditions, interactions can reduce overall efficiency due to adverse changes in coherent structures.

Kouser et al. reviewed such hybrid systems, emphasizing the importance of surface design, wettability and roughness factors.[15] These studies highlight that combining mechanisms is not automatically beneficial; coherent-structure interactions must be carefully considered.

2.9.2 Other Hybrid Concepts

Other hybrid concepts include:

- Riblet–superhydrophobic composites, which combine geometric suppression of cross-flow with slip.
- Porous walls with spanwise forcing, combining permeability and active transverse motion.
- Multi-scale textured surfaces that simultaneously target different coherent structures across scales.[2, 5]

These strategies demonstrate the richness of coherent-structure-based design but also underline the risk of unintentionally promoting new drag-producing structures.

2.10 Synthesis and Relevance to Passive Microgrooves

Across passive, active and hybrid control strategies, several common themes emerge:

- **Near-wall coherent structures are central.** Drag reduction is consistently associated with weakening or reorganizing quasi-streamwise vortices, streaks and hairpins in the buffer layer.[3, 4]

- **Geometry and slip must be matched to viscous scales.** Riblets and microgrooves perform best when their height and spacing are tuned to viscous-sublayer thickness, while SHSs and LISs require slip lengths appropriate to near-wall structure sizes.[2, 5–7]
- **Permeability and multi-phase interfaces are double-edged tools.** Porous walls and multi-phase surfaces can significantly reduce drag but can also generate new coherent structures that increase drag if design thresholds are exceeded.[8, 9, 15]
- **Actuation strategies are most effective when aligned with coherent-structure dynamics.** Spanwise forcing and DRL-based control work by targeting specific events (ejections, sweeps) and structures (vortices, streaks) in time and space.[10, 11, 13, 17]

In the context of this thesis, passive microgroove surfaces in a rectangular turbulent channel are designed and evaluated with these insights in mind. Microgroove geometries are chosen to interact with near-wall coherent structures in ways analogous to riblets, and their drag-reduction performance is interpreted using the coherent-structure framework developed in this chapter. The Methodology chapter will detail the specific designs, flow conditions and measurement techniques, while subsequent chapters will analyze how these surfaces perform relative to the mechanisms summarized here.

Chapter 3

Research Methodology

3.1 Overview

This chapter describes the methodology used to investigate drag reduction via coherent-structure modification in a turbulent rectangular water channel equipped with passive microgroove surfaces. The focus is on how the experimental facility, microgroove geometries, measurement procedures and data-reduction steps are designed to allow physically consistent interpretation using turbulent channel-flow theory and coherent-structure concepts. Where possible, established hydraulic and turbulence relations are adopted, and their applicability to the present geometry and Reynolds-number range is discussed.

The main elements of the methodology are:

- Definition and characterization of the rectangular test channel and working fluid.
- Hydraulic calculations for Reynolds number, friction velocity, viscous-sub-layer thickness and friction factor.
- Design and fabrication of microgroove surfaces based on viscous-sublayer scaling, microfabrication constraints and shark-skin-inspired textures.
- Measurement of volumetric flow rate and differential pressure, and conversion to friction factors and drag-reduction metrics.
- Use of friction-factor correlations and the Fukagata–Iwamoto–Kasagi (FIK) identity to connect measured drag changes to near-wall coherent structures.
- Uncertainty analysis of key measured and derived quantities.

3.2 Experimental Facility and Working Fluid

3.2.1 Rectangular Channel Geometry

The experiments are conducted in a straight, horizontal, closed water channel fabricated from transparent acrylic to allow optical access. The test section has a rectangular cross-section of width $w = 51.2$ mm and height $h = 6.4$ mm, corresponding to a cross-sectional area

$$A = wh,$$

and a wetted perimeter

$$P = 2(w + h).$$

For a non-circular duct, the hydraulic diameter is defined as[1, 27]

$$D_h = \frac{4A}{P}.$$

With the present dimensions, this gives

$$D_h \approx 11.4 \text{ mm}.$$

The channel aspect ratio is

$$\Gamma = \frac{w}{h} \approx 8.$$

High-aspect-ratio rectangular ducts are known to exhibit friction factors that deviate from those of equivalent circular pipes due to modified velocity distributions and increased wetted perimeter.[27, 28] This motivates the use of a hydraulic-diameter-based Reynolds number combined with aspect-ratio corrections in friction-factor correlations, as discussed in Section 3.3.

3.2.2 Flow Loop and Operating Conditions

The channel is integrated into a closed-loop flow system consisting of a reservoir, pump, flow-conditioning section, test section and return line. Water at a controlled temperature of 30 °C is used as the working fluid.

Bulk flow velocity U_b is set by adjusting pump speed and measured from volumetric flow rate Q via

$$U_b = \frac{Q}{A}.$$

Volumetric flow rates are chosen such that the bulk velocity U_b varies from approximately 0.51 to 1.19 m/s, yielding hydraulic-diameter Reynolds numbers

$$Re_{D_h} = \frac{U_b D_h}{\nu},$$

in the turbulent range $Re_{D_h} \approx 7.2 \times 10^3 - 1.7 \times 10^4$, where ν is the kinematic viscosity of water at 30 °C. These Reynolds numbers span transitional and fully turbulent regimes for internal flows in smooth ducts, and are sufficiently high that coherent structures in the buffer and logarithmic regions are well developed.

The test section length L is chosen such that the flow entering the measurement region is fully developed for each operating condition. Entry-length correlations for rectangular ducts are used to confirm that turbulent development lengths are smaller than the distance from the inlet to the first pressure tap.[27]

3.3 Hydraulic Characterization

3.3.1 Reynolds Number and Flow Regime

The hydraulic Reynolds number Re_{D_h} is used to classify the flow regime, as defined above. For smooth pipes and ducts, the onset of turbulence typically occurs near $Re_{D_h} \approx 2,000 - 3,000$, with fully turbulent regimes at $Re_{D_h} \gtrsim 4,000$. [1] The present experiments principally operate in the turbulent regime, although some low-flow cases may fall into transition. Data from low-Reynolds-number conditions are treated cautiously and, where appropriate, excluded from coherent-structure-based interpretations that assume fully developed turbulence.

3.3.2 Fanning Friction Factor and Pressure Drop

The Fanning friction factor for internal flows is defined as [29]

$$f = \frac{\tau_w}{\frac{1}{2}\rho U_b^2},$$

where τ_w is the mean wall shear stress and ρ is the fluid density. For a straight duct of length L , the pressure drop Δp associated with friction can be related to f via the

standard internal-flow relation

$$\Delta p = 4f \left(\frac{\rho U_b^2}{2} \right) \frac{L}{D_h}.$$

This equation is equivalent to the Darcy–Weisbach formulation if the Darcy friction factor f_D is related to the Fanning friction factor by $f_D = 4f$.[\[30\]](#)

Solving for f yields

$$f = \frac{\Delta p D_h}{2\rho U_b^2 L}.$$

In the experiments, Δp is measured between two pressure taps separated by a known distance L in the fully developed region of the test section. Bulk velocity U_b is inferred from flow-rate measurements as described above. Using these measurements and known geometric quantities, Fanning friction factors are computed for each surface configuration and operating condition.

3.3.3 Baseline Friction-Factor Correlations

For hydraulically smooth circular pipes, the classical Blasius correlation for turbulent friction factor is[\[31, 32\]](#)

$$f_{\text{Blasius}} = 0.079 Re^{-0.25},$$

where f is the Fanning friction factor and Re is defined using pipe diameter and bulk velocity. This correlation is valid approximately for $4 \times 10^3 \lesssim Re \lesssim 10^5$ and is widely used as a baseline for smooth turbulent flows.[\[29\]](#)

In rectangular channels, friction factors differ due to aspect ratio effects. Semi-empirical correlations and numerical studies show that friction factors for rectangular channels can be approximated by correcting circular-pipe correlations with aspect-ratio-dependent factors.[\[27, 28\]](#) For example, Davidson presents correlations in which a Blasius-type base friction factor is multiplied by a polynomial correction factor in aspect ratio.[\[27\]](#)

In this thesis, smooth-channel friction factors for the acrylic reference plate are compared against Blasius-based estimates and, where appropriate, against rectangular-channel correlations to validate measurement accuracy and to identify deviations due to aspect ratio and finite-length effects.

3.3.4 Wall Shear Stress and Friction Velocity

Once the Fanning friction factor f is known, the mean wall shear stress τ_w can be obtained from its definition:

$$\tau_w = f \left(\frac{1}{2} \rho U_b^2 \right).$$

The friction velocity is defined as

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}}.$$

Using this relation, measured pressure drops and derived friction factors can be converted into friction velocities for each operating condition and surface configuration.

The friction velocity provides a natural velocity scale for wall-bounded turbulence and is used to define viscous wall units:

$$y^+ = \frac{y u_\tau}{\nu}, \quad u^+ = \frac{u}{u_\tau},$$

where y is the wall-normal coordinate and u is streamwise velocity.[3]

3.3.5 Viscous Sub-Layer Thickness and Microgroove Scaling

The viscous sublayer in turbulent wall-bounded flow is typically characterized by $y^+ \lesssim 5$, where viscous effects dominate and the flow is nearly linear in y . [3] An approximate physical thickness for the viscous sublayer can therefore be defined as

$$\delta_v \approx 5 \frac{\nu}{u_\tau}.$$

This thickness is used as a design guide for microgroove height. To ensure that microgroove tips lie within the viscous sublayer at the operating Reynolds numbers, their physical height h_g is chosen such that

$$h_g \lesssim \delta_v.$$

Using measured or estimated friction velocities for each operating condition, viscous-sublayer thicknesses are computed and used to assess microgroove designs. This scaling is consistent with the riblet literature, where optimal riblet heights are typically of order 6–15 wall units, i.e. within or just above the viscous sublayer.[2, 5]

3.4 Microgroove Surface Design and Fabrication

3.4.1 Design Objectives

Microgroove surfaces are designed to:

- Interact primarily with near-wall coherent structures (quasi-streamwise vortices and streaks) within the viscous sublayer and buffer layer.
- Exploit geometric scaling in viscous units to produce drag reduction without introducing large-scale adverse structures such as Kelvin–Helmholtz rollers.
- Be manufacturable using available 3D printing technology (Elegoo Saturn 4 Ultra 16 K), considering minimum printable feature height and tolerances.
- Be robust in the water channel environment, with minimal deformation or damage under operating conditions.

3.4.2 Shark-Skin-Inspired Denticle Geometry

One of the most prominent classes of bio-inspired microstructures investigated in this research is shark-skin-inspired dermal denticles. The baseline geometric model was adapted from an open-source biological surface scan. Due to slight biological asymmetries in the raw scan, the geometry was refined by mirroring one half across the longitudinal plane of symmetry, ensuring a highly symmetric and reproducible shape. Furthermore, a planar substrate was added beneath the denticle base to facilitate the definition of geometric mates and enable the generation of regular linear patterns in SolidWorks.

The comprehensive engineering drawing, standard orthographic views, and precise dimensions of this 3D model, along with the detailed ridges on the crown, are presented in Figure 3.1.

To systematically investigate geometric scale effects, four distinct configurations were established based on geometric scaling factors:

1. **Full Scale (1.00x):** The complete original scale of the biological denticle, retaining all organic 3D contours and longitudinal riblet ridges at full scale.
2. **Subscale Model (0.30x):** A miniaturized version aimed at reducing protrusion height to confine the denticles within the viscous sublayer.

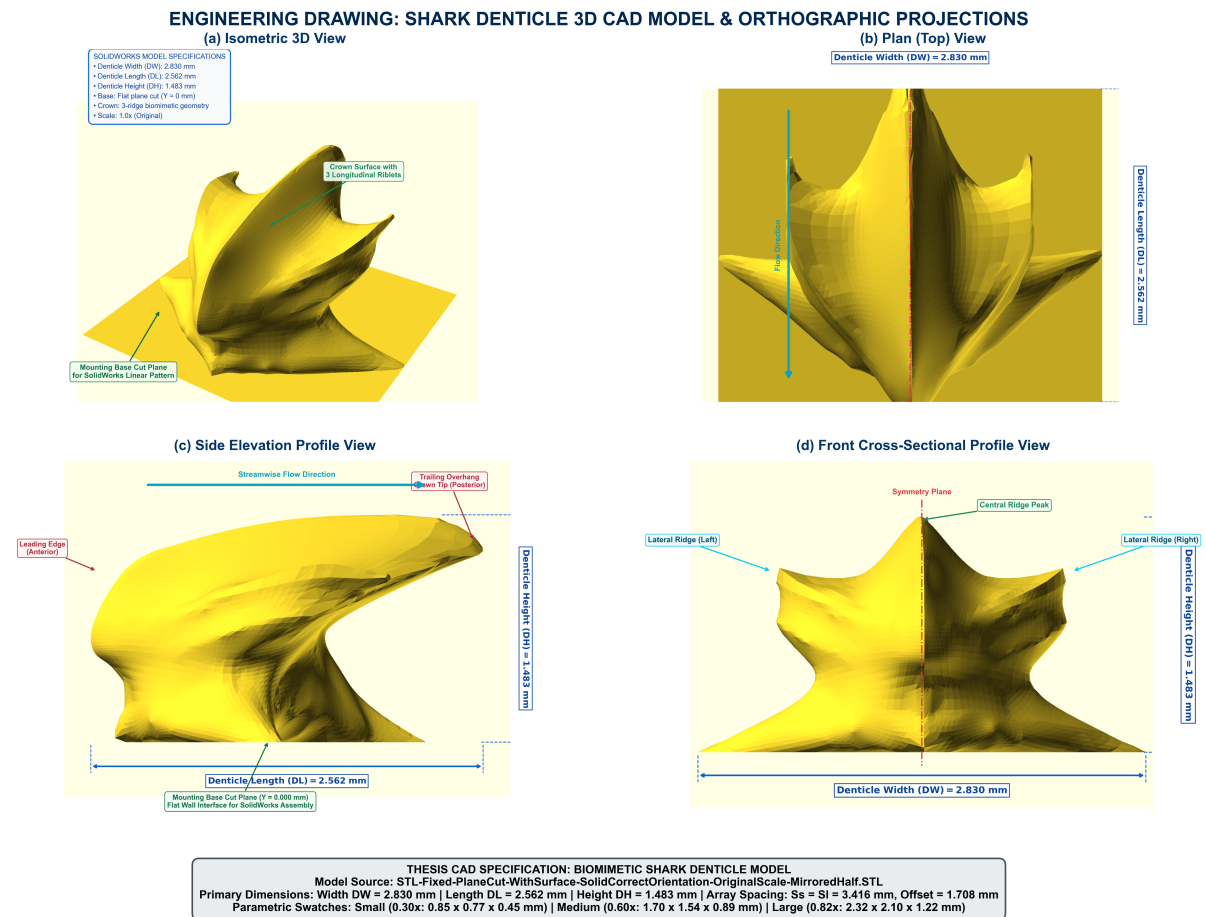


Figure 3.1: Engineering drawing and orthographic projections of the 3D shark denticle model in SolidWorks at full original scale (1.00x): (a) 3D isometric view showing the tri-riblet crown and planar trimmed mounting base, (b) top view (planform) with crown width 2.830 mm, length 2.562 mm, axis of symmetry, and flow direction, (c) side view showing overall height 1.483 mm, trailing-edge tip elevation above the substrate, and leading-edge profile, and (d) front view showing central peak elevation and lateral riblets symmetric about the centerline.

3. **Intermediate Model (0.60x):** A medium scale designed to evaluate transition behavior and aspect-ratio effects.
4. **Enlarged Model (0.82x):** An intermediate enlarged scale with lower packing density and wider spacing.

The geometric dimensions and array layout parameters for these four configurations across the active plate area (width 47.0 mm, length 165.0 mm) are summarized in Table 3.1. The spatial staggered layout and array spacings between denticles are illustrated in Figure 3.2.

Table 3.1: Geometric dimensions and array layout parameters for different shark denticle configurations

Design Parameter (mm)	Full Scale (1.00x)	Subscale (0.30x)	Intermediate (0.60x)	Enlarged (0.82x)
Scaling Factor	1.00	0.30	0.60	0.82
Denticle Length (DL)	2.562	0.769	1.537	2.101
Denticle Width (DW)	2.830	0.849	1.698	2.321
Denticle Height (DH)	1.483	0.445	0.890	1.216
Streamwise Spacing (D_1 or S_s)	3.416	0.800	2.050	4.100
Spanwise Spacing (D_3 or S_l)	3.416	0.800	2.050	4.100
Staggered Row Offset (D_2)	1.708	0.400	1.025	2.050
Active Array Width	47.0	47.0	47.0	47.0
Active Array Length	165.0	165.0	165.0	165.0
Total Number of Rows (N_{rows})	97	413	161	81
Number of Columns per Row (N_{cols})	13	59	23	12
Total Denticle Count per Plate	1,261	24,367	3,703	972

Fabrication Constraints and Infeasibility of Subscale Denticles

A comprehensive feasibility assessment using the stereolithography 3D printer (Elegoo Saturn 4 Ultra 16K) demonstrated that the subscale model (0.30x) and intermediate configurations (0.60x and 0.82x) could not be reliably printed, post-processed, or experimentally deployed due to physical and chemical photopolymerization constraints, as well as hydrodynamic incompatibility. The underlying technical reasons are:

1. **Photopolymerization Resolution Limits and Negative Undercuts:** Although the nominal XY pixel pitch of the printer is $19\ \mu\text{m}$, UV light scattering and refraction (optical bleeding) inevitably diffuse into neighboring resin regions. Natural shark denticles exhibit complex 3D overhangs where the elevated crown cantilevers over a

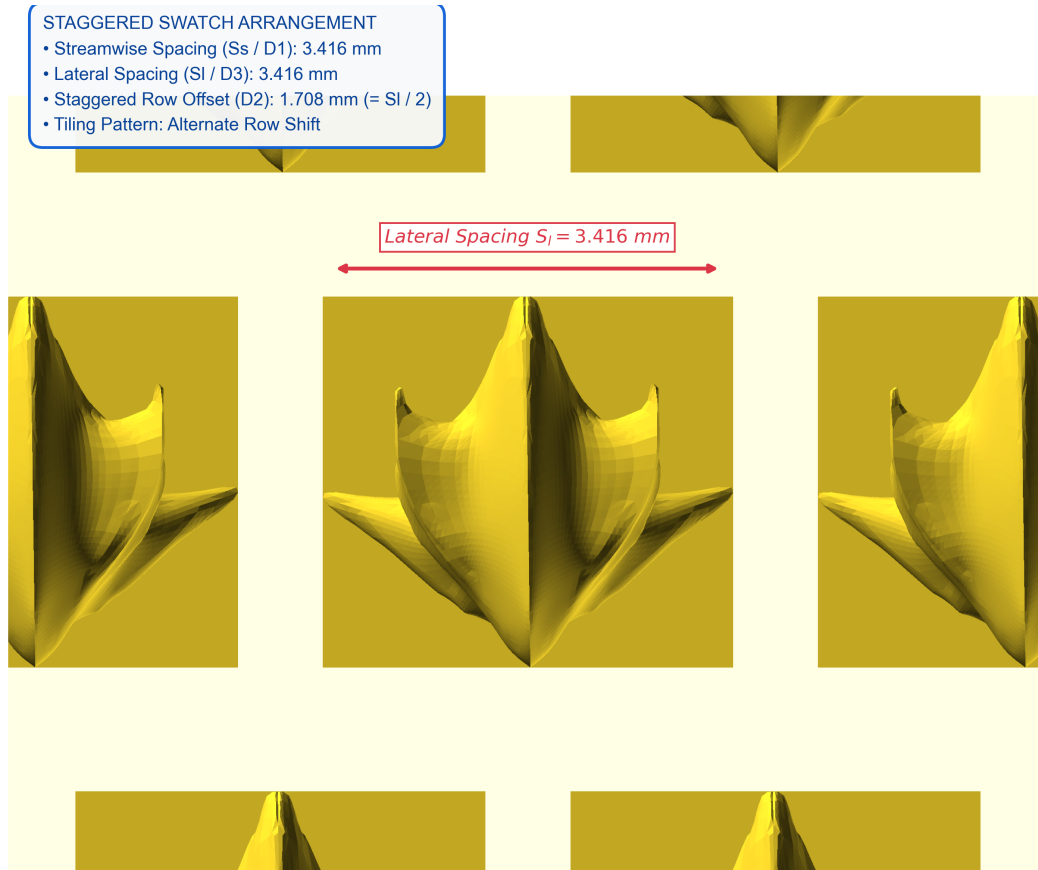


Figure 3.2: Staggered array layout of shark denticles on the test plate, illustrating streamwise pitch ($S_s = 3.416 \text{ mm}$), spanwise pitch ($S_l = 3.416 \text{ mm}$), and the alternating row offset ($D_2 = 1.708 \text{ mm}$).

narrower stem base. At 0.30x scale (length 0.769 mm, height 0.445 mm), trailing-edge features and crown micro-ridges possess thicknesses of less than 30–50 μm . These micro-cavities become cured and clogged due to optical bleeding of trapped uncured resin, obliterating geometric fidelity.

2. **Capillary Resin Trapping and Liquid Bridging:** In the subscale (0.30x) specimen, 24 367 miniature denticles are packed at an ultra-dense pitch of 0.800 mm. Given a denticle width of 0.849 mm, the clearance between adjacent crowns in the staggered layout drops below 50 μm . High surface tension and liquid resin viscosity prevent gravity drainage between successive exposure layers. Uncured liquid resin remains trapped within the inter-denticle gaps and is subsequently polymerized by stray UV light, fusing the array into a distorted, quasi-solid polymer block.
3. **FEP Peeling Forces and Support Structure Limitations:** In bottom-up SLA printing, after each layer is exposed, the build platform is retracted upward to peel the cured layer from the FEP release film, generating significant suction peeling forces. For 0.30x denticles with 24 367 slender stems, these forces induce severe shear yielding, twisting, or total detachment from the substrate. Furthermore, generating slicer support structures for sub-millimeter features (0.4 mm) is impossible, and even if generated, physical manual support removal from underneath 24,000 denticles without structural destruction is utterly infeasible.
4. **Mechanical Damage during Ultrasonic Post-Processing and Drying:** To eliminate uncured resin residue, parts must undergo ultrasonic agitation in high-purity isopropyl alcohol (IPA) followed by high-pressure air drying. The energetic acoustic cavitation waves and high-momentum air jets reliably shatter the brittle subscale stems, shearing denticles off the substrate.
5. **Hydrodynamic Incompatibility of Intermediate Scales with the Viscous Sublayer:** Beyond resin trapping challenges in the 0.60x (3703 denticles) and 0.82x (972 denticles) models, hydraulic calculations demonstrated that the viscous sublayer thickness at operating velocities is only 0.056 mm to 0.118 mm. In contrast, the denticle heights are 0.890 mm (0.60x) and 1.216 mm (0.82x)—exceeding the sublayer thickness by factors of 8 to 21 ($y^+ \approx 15\text{--}60$). At these elevations, the denticles act as bluff obstructions, triggering boundary-layer separation, horseshoe vortex shedding, and severe form drag. Consequently, fabricating these intermediate configurations offered neither structural feasibility nor hydrodynamic utility.

Fabrication of Full-Scale (1.0x) Denticles and Physical Origins of Drag Increase

Unlike the miniaturized models, the full-scale (1.00x) denticle configuration (height 1.483 mm, width 2.830 mm, open spacing 3.416 mm) was fully compatible with the manufacturing envelope of the Elegoo Saturn 4 Ultra 16 K. Millimeter-scale clearances permitted free drainage of liquid resin without capillary bridging, and the robust stems easily endured FEP peeling and ultrasonic washing. A full-scale test plate featuring 1261 denticles in a staggered array was successfully printed, post-processed, and tested in the water channel.

However, as detailed in Chapter 4 (Table 4.2), differential pressure measurements revealed that this surface produced a **substantial drag increase** across all operating flow rates:

- At $Q = 0.6 \text{ m}^3/\text{h}$, pressure drop reached 1.30 cmH₂O, representing a drag increase of 12.3 % relative to the aluminum base and 18.2 % relative to smooth acrylic.
- At $Q = 1.0 \text{ m}^3/\text{h}$, pressure drop surged to 4.30 cmH₂O, corresponding to a severe drag penalty of 32.2 % (aluminum base) and 43.3 % (acrylic reference).
- At $Q = 1.4 \text{ m}^3/\text{h}$, pressure drop reached 7.00 cmH₂O, indicating a drag increase of 27.7 % (aluminum) and 28.3 % (acrylic).

The physical mechanism driving this drag penalty stems directly from violating the fundamental viscous scaling requirement ($h_d^+ < 5$):

1. **Deep Penetration into Buffer and Logarithmic Layers ($y^+ \gg 5$):** The full denticle height ($h_d = 1.483 \text{ mm}$) is 13 to 26 times thicker than the viscous sublayer ($\delta_v \approx 0.056\text{--}0.118 \text{ mm}$), protruding to $y^+ \approx 15\text{--}30$ and beyond. The structures emerge from the protective viscous sublayer and directly encounter high-speed turbulent buffer and log-law flow.
2. **Bluff-Body Aerodynamics and Wake Shedding:** Each macroscopic denticle behaves as a three-dimensional bluff obstacle. At the leading edge, the oncoming boundary layer stagnates and rolls up into energetic horseshoe vortices; severe lateral and trailing-edge separation ejects turbulent hairpin packets into the channel core.
3. **Direct Wake Impingement in Staggered Layout:** In the alternating staggered arrangement, turbulent wakes shed from upstream denticles impinge directly upon the frontal crowns of downstream denticles, dramatically amplifying pressure fluctuations and turbulent kinetic energy dissipation.

4. Overwhelming Dominance of Form Drag over Skin-Friction Reduction:

The static pressure differential between the high-pressure stagnation zone ahead of the denticle and the low-pressure suction wake behind it generates massive form drag, entirely overwhelming any localized frictional benefits.

These findings prove that bio-inspired microstructures cannot achieve drag reduction without strict adherence to viscous inner scaling ($h^+ < 5$). This realization guided the subsequent optimization toward truncated "tip-only" profiles and, crucially, continuous longitudinal microgrooves, which successfully achieved over 22% net drag reduction.

3.4.3 Engineering Profiles of Longitudinal Microgrooves (Triangular, Rectangular, and Trapezoidal)

In addition to denticles, regular longitudinal microgroove geometries were designed and fabricated as passive turbulence control surfaces operating within the viscous sublayer. Oriented continuously along the mean flow direction, these grooves restrict spanwise crossflow fluctuations (w' and v') while pinning and dampening quasi-streamwise vortices.

Four distinct microgroove cross-sections modeled in OpenSCAD are depicted in Figure 3.3:

1. **Symmetric Triangular Longitudinal Riblets (Figure 3.3a):** Adapted from classical Bechert designs with optimal pitch $s = 0.43 \text{ mm}$ ($430 \mu\text{m}$), peak height $h = 0.37 \text{ mm}$ ($370 \mu\text{m}$), sharp apex angle $2\alpha = 60^\circ$, and height-to-pitch ratio $h/s = 0.866$. At $U_b = 0.40 \text{ m/s}$, these dimensions correspond to $s^+ \approx 15$ and $h^+ \approx 13$ near the upper edge of the viscous sublayer.
2. **Superhydrophobic Rectangular Microgrooves (Figure 3.3b):** Featuring blade width $w = 45 \mu\text{m}$, groove opening width $g = 35 \mu\text{m}$, pitch $p = w + g = 80 \mu\text{m}$, and depth $h = 100 \mu\text{m}$ (matching two $50 \mu\text{m}$ SLA print layers). Vertical sidewalls ($\theta = 0^\circ$) yield an aspect ratio $h/g = 2.86$, solid fraction $\phi_s = 56.25\%$, and open gas fraction $\phi_g = 43.75\%$. Under two-phase flow, the water–air meniscus suspends across the tips in a stable Cassie–Baxter plastron state.
3. **Trapezoidal Microgrooves with Positive Draft Angle ($+5^\circ$) (Figure 3.3c):** To enhance structural blade stiffness and facilitate resin drainage, sidewalls converge toward the trench floor with positive draft angle $\theta = +5^\circ$. Maintaining top dimensions ($w_{\text{top}} = 45 \mu\text{m}$, $g_{\text{top}} = 35 \mu\text{m}$), the blade base widens to $w_{\text{bot}} = 62.5 \mu\text{m}$, narrowing bottom groove width to $g_{\text{bot}} = 17.5 \mu\text{m}$.

4. **Re-entrant Trapezoidal Microgrooves with Negative Draft Angle (-5°)** (Figure 3.3d): Designed to maximize capillary plastron stability against turbulent pressure pulses, sidewalls diverge with negative angle $\theta = -5^\circ$, forming an overhanging re-entrant lip ($g_{\text{top}} = 35 \mu\text{m}$) while expanding the trench floor to $g_{\text{bot}} = 52.5 \mu\text{m}$ with narrow base width $w_{\text{bot}} = 27.5 \mu\text{m}$.

Cross-Sectional Morphology and Detailed Engineering Dimensions of 3D-Printed Micro-Surfaces

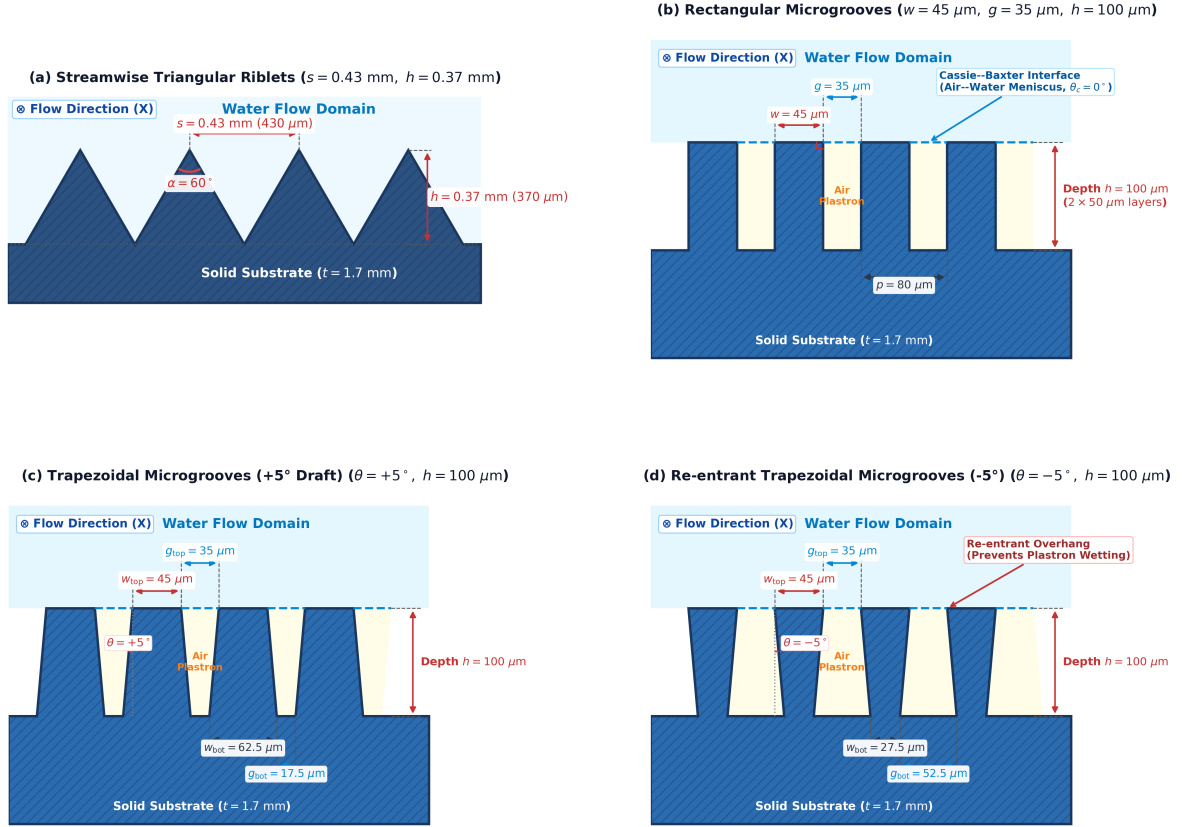


Figure 3.3: Engineering front-view cross-sections and detailed dimensions of printed surface microstructures: (a) symmetric triangular riblets with pitch 0.43 mm and height 0.37 mm , (b) superhydrophobic rectangular microgrooves ($w = 45 \mu\text{m}$, $g = 35 \mu\text{m}$, $p = 80 \mu\text{m}$, $h = 100 \mu\text{m}$) with Cassie–Baxter interface, (c) trapezoidal microgrooves with positive draft angle ($+5^\circ$) and base reinforcement ($w_{\text{bot}} = 62.5 \mu\text{m}$), and (d) re-entrant trapezoidal microgrooves (-5°) with overhang lips and expanded cavity base ($g_{\text{bot}} = 52.5 \mu\text{m}$). All microstructures rest upon a solid 1.7 mm photopolymer substrate.

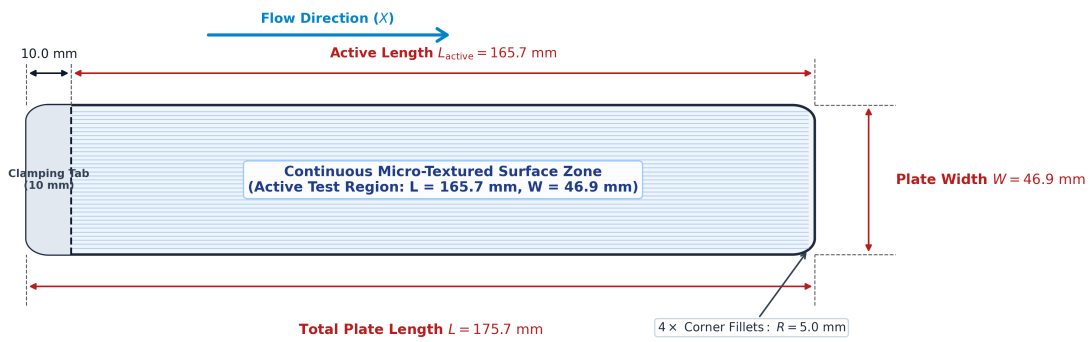
Perpendicular microgrooves (oriented 90° to the mean flow) were included as a contrasting baseline to confirm that misaligned surface textures generate severe secondary eddy motions and substantial drag increases.

3.4.4 Test Plate Fabrication and Dimensions

Surface specimens were fabricated using the Elegoo Saturn 4 Ultra 16K SLA printer. The complete engineering drawing of the test plate is presented in Figure 3.4. The plate has an overall length of 175.7 mm, width of 46.9 mm, upstream clamping tab length of 10.0 mm, active micro-textured length of 165.7 mm, and corner fillets with radius 5.0 mm for flush sealing within the channel bottom recess. The rigid photopolymer base thickness of 1.7 mm ensures flexural rigidity under operational hydrodynamic loads.

Engineering Drawing of 3D-Printed Experimental Test Plate (Top & Front Views)

(a) Top View (Orthographic Projection of Test Plate)



(b) Front View (Cross-Sectional Thickness and Width)

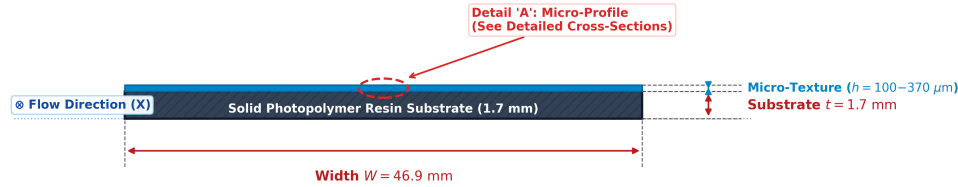


Figure 3.4: Two-view engineering drawing of the SLA 3D-printed test plate: (a) top planform view indicating overall length (175.7 mm), outer width (46.9 mm), clamping tab (10.0 mm), active textured length (165.7 mm), and corner radii (5.0 mm); (b) front elevation showing solid photopolymer base thickness (1.7 mm) and surface microstructure layer position (Detail 'A').

Key fabrication steps included:

- Slicing with fine layer resolution ($50 \text{ }\mu\text{m}$) and calibrated UV exposure to resolve micron-scale tip features.
- Optimized build-volume orientation to prevent support scarring across the active patterned region.
- Rigorous multi-stage chemical washing and photochemical post-curing.
- Precision flatness inspection to eliminate mounting misalignments.

3.5 Instrumentation and Data Acquisition

3.5.1 Volumetric Flow-Rate Measurement

Volumetric flow rate Q is measured using a calibrated flowmeter (e.g. turbine or electromagnetic) installed upstream of the test section. Flowmeter calibration curves provided by the manufacturer, or obtained by comparison with gravimetric measurements, are used to convert instrument readings (e.g. pulses or voltage) into volumetric flow rate with known uncertainty.

Bulk velocity U_b is then computed as

$$U_b = \frac{Q}{A},$$

for each measurement point. This velocity is used in Reynolds-number and friction-factor calculations.

3.5.2 Differential Pressure Measurement

Wall friction is inferred from pressure-drop measurements using differential pressure transducers. Two pressure taps are installed at the top or side wall of the channel, separated by a known distance L within the fully developed region of the flow.

The transducer output is converted to pressure difference Δp via calibration curves. Sampling is performed at a rate sufficient to capture slow temporal fluctuations in turbulent pressure; time averaging is used to obtain mean pressure drops for each operating condition and surface configuration.

The Fanning friction factor is then computed from

$$f = \frac{\Delta p D_h}{2\rho U_b^2 L}.$$

3.5.3 Temperature and Fluid Properties

Fluid temperature is monitored using thermocouples or RTDs in the reservoir and at multiple locations in the loop. Assuming the flow is nearly isothermal, the mean temperature is used to determine fluid properties such as density ρ and kinematic viscosity ν from standard water-property tables.

Temperature data are also used to assess whether changes in viscosity or density

between runs could influence friction-factor measurements. When necessary, friction factors are corrected for small property variations.

3.6 Data Reduction and Drag-Reduction Metrics

3.6.1 Reynolds Number and Friction Factor for Each Surface

For each combination of surface configuration and volumetric flow rate:

1. Compute bulk velocity U_b from measured Q .
2. Compute Reynolds number $Re_{D_h} = U_b D_h / \nu$.
3. Compute friction factor f from measured Δp using the relation above.
4. Compute wall shear stress $\tau_w = f(\frac{1}{2}\rho U_b^2)$ and friction velocity $u_\tau = \sqrt{\tau_w / \rho}$.
5. Compute viscous-sublayer thickness $\delta_v \approx 5\nu / u_\tau$.

These steps are applied to both smooth reference surfaces (aluminum base plate and smooth acrylic plate) and microgroove surfaces.

3.6.2 Definition of Drag-Reduction Percentage

Drag reduction is quantified in terms of the relative change in friction factor or pressure drop between the microgroove surface and a chosen baseline surface at the same bulk Reynolds number. Using the Fanning friction factor, the drag-reduction percentage is defined as

$$DR_f = \frac{f_{\text{baseline}} - f_{\text{microgroove}}}{f_{\text{baseline}}} \times 100\%.$$

Alternatively, using pressure drop,

$$DR_{\Delta p} = \frac{\Delta p_{\text{baseline}} - \Delta p_{\text{microgroove}}}{\Delta p_{\text{baseline}}} \times 100\%.$$

In practice, both definitions are equivalent when comparing surfaces at fixed Re_{D_h} . Percentages are reported for each flow rate and surface, allowing direct comparison of drag reduction across geometries.

3.6.3 Baseline Selection

Two baselines are considered:

- A smooth aluminum base plate representing a conventional duct wall.
- A smooth acrylic plate representing the same material as the microgroove surfaces but without microstructure.

Comparing microgroove surfaces to both baselines provides insight into:

- Net drag reduction relative to conventional metallic walls.
- Incremental drag reduction attributable to microgrooves beyond any changes associated with material substitution.

3.7 Relations to Coherent Structure Analysis

3.7.1 Fukagata–Iwamoto–Kasagi Identity

To relate measured drag changes to coherent structures, the Fukagata–Iwamoto–Kasagi (FIK) identity is introduced as a theoretical framework.^[33] For fully developed turbulent channel flow, the skin-friction coefficient can be decomposed as

$$C_f = \frac{12}{Re_b} + 12 \int_0^1 2(1-y) [-\langle u'v' \rangle] dy + (\text{inhomogeneous and transient terms}),$$

where Re_b is the bulk Reynolds number, y is the normalized wall-normal coordinate, $\langle u'v' \rangle$ is the Reynolds shear stress, and the final term accounts for inhomogeneous and transient effects.^[33]

In the context of steady, fully developed channel flow without external forcing, this simplifies to

$$C_f \approx \frac{12}{Re_b} + 12 \int_0^1 2(1-y) [-\langle u'v' \rangle] dy.$$

This expression shows that turbulent contributions to skin friction arise from the distribution of Reynolds shear stress, which is directly linked to coherent structures such as ejections and sweeps.

While the present experiments do not measure velocity fields directly, the FIK identity provides a conceptual bridge between measured friction factors and coherent-structure behavior:

- Drag reduction corresponds to a reduction in the integral of Reynolds shear stress, implying changes in coherent structures.
- Passive microgrooves can be interpreted as devices that alter $\langle u'v' \rangle$ in the near-wall region by reshaping vortices and streaks.

3.7.2 Interpretive Framework

The methodology therefore supports the following interpretive steps:

1. Use measured friction factors and drag-reduction percentages to infer changes in overall skin friction.
2. Use viscous-sublayer thickness and microgroove scaling to argue that the primary impact of microgrooves is on near-wall coherent structures.
3. Use literature on riblets, SHSs, porous walls and polymer drag reduction to qualitatively connect measured drag changes to expected modifications in coherent structures, as detailed in the Related and Past Works chapter.

The combination of experimental data and theoretical frameworks (Blasius-type friction-factor correlations, rectangular-channel corrections and the FIK identity) enables a coherent-structure-based interpretation of passive microgroove performance.

3.8 Uncertainty Analysis

3.8.1 Measurement Uncertainties

Uncertainties in volumetric flow rate, pressure drop and temperature propagate into uncertainties in derived quantities such as Reynolds number, friction factor and drag-reduction percentage.

Representative sources of uncertainty include:

- Flowmeter calibration errors and resolution limits.
- Differential pressure transducer calibration errors, zero offsets and drift.
- Geometric measurement errors in channel dimensions (w , h , L), particularly if surfaces are replaced or re-mounted.

- Temperature measurement errors affecting fluid-property estimates.

Standard uncertainty analysis techniques are applied, typically using linear propagation of errors:

$$u_f \approx \sqrt{\left(\frac{\partial f}{\partial \Delta p} u_{\Delta p}\right)^2 + \left(\frac{\partial f}{\partial U_b} u_{U_b}\right)^2 + \left(\frac{\partial f}{\partial D_h} u_{D_h}\right)^2 + \left(\frac{\partial f}{\partial L} u_L\right)^2},$$

where $u_{(\cdot)}$ denotes the standard uncertainty of the corresponding quantity. Similar expressions are used for Re_{D_h} and DR .

3.8.2 Repeatability and Statistical Uncertainty

To quantify repeatability, multiple runs are performed for selected operating conditions and surfaces. Mean values and standard deviations of f , Re_{D_h} and DR are computed across runs, and statistical uncertainty is estimated.

Where possible, confidence intervals (e.g. 95% intervals) are reported. These intervals provide bounds within which true drag-reduction values are expected to lie, accounting for measurement scatter.

3.9 Summary

This chapter has described the methodology used to characterize drag reduction by passive microgroove surfaces in a turbulent rectangular water channel. Key elements include hydraulic definitions (Reynolds number, friction factor, friction velocity), viscous-sublayer-based microgroove design, differential-pressure and flow-rate measurement procedures, and theoretical frameworks for interpreting skin-friction changes in terms of coherent structures.

The methodology is designed to support a coherent-structure-based analysis of drag reduction: geometric and operating parameters are chosen to ensure that microgrooves primarily interact with near-wall structures, while measurement and data-reduction procedures yield reliable friction-factor and drag-reduction metrics that can be compared against theoretical and DNS-based insights from the literature.

3.10 Post-Processing of 3D-Printed Microgroove and Denticle Surfaces

High-resolution stereolithography (SLA) printing leaves a thin layer of uncured resin on surfaces and, critically for microgroove geometries, trapped resin inside narrow gaps and cavities.[34, 35] If this residue is not removed, it can partially or completely block microgrooves, alter the designed cross-section, and change the effective roughness and hydraulic diameter, thereby invalidating the assumed geometry used in viscous scaling and coherent-structure interpretations.[36] For this reason, a dedicated post-processing protocol is used to clean and fully cure the printed surfaces before installation in the water channel.

3.10.1 Immediate Immersion in High-Purity Alcohol

Immediately after printing and removing the build plate from the printer, the microgroove or denticle surface is detached and placed into a bath of high-purity isopropyl alcohol (IPA) at 99.8% concentration. High-purity IPA is widely recommended for resin-print cleaning because it efficiently dissolves uncured resin while minimizing water content, which can reduce cleaning effectiveness and interfere with subsequent UV curing.[34, 37, 38] Immediate immersion ensures that resin still in a liquid or gel state is dissolved before it can partially cure in gaps or channels.

The initial IPA bath serves several purposes:

- It dissolves uncured resin from external surfaces and from within microgroove cavities.
- It reduces the viscosity of trapped resin, making it easier to dislodge during ultrasonic cleaning.
- It prevents early ambient-light curing of resin inside gaps, which would be difficult to remove mechanically.

3.10.2 Ultrasonic Cleaning for 10 Minutes

After immersion, the printed model is placed in an ultrasonic cleaning device filled with the same high-purity IPA and cleaned for approximately 10 minutes. Ultrasonic cleaning uses high-frequency acoustic waves to generate cavitation and microstreaming in the

solvent, which is particularly effective at dislodging uncured resin from narrow channels and complex microstructures.[35, 37]

In the context of microgrooves and shark-skin denticles:

- Ultrasonic agitation drives IPA into the grooves and between denticles, dislodging resin that simple soaking cannot reach.
- The 10-minute duration balances thorough cleaning with avoidance of over-exposure to IPA, which can swell or weaken some resins if applied for excessively long times.[39, 40]
- Cleaning in IPA at room temperature avoids thermal effects that could deform fine structures.

Similar ultrasonic cleaning times (5–30 minutes) have been reported in microfluidic and biomicrofluidic fabrication for clearing sub-100 μm channels of uncured resin prior to UV curing and heat treatment.[34, 35]

3.10.3 High-Pressure Air Blowing to Remove IPA

Following ultrasonic cleaning, the printed surface is removed from the IPA bath and subjected to high-pressure air blowing using a compressed-air source. This step serves to:

- Physically expel IPA and any remaining resin fragments from microgroove gaps and denticle cavities.
- Promote rapid drying of residual solvent on external surfaces.
- Prevent trapped IPA from inhibiting subsequent UV curing, as residual solvent can shield resin from UV exposure and lead to uneven curing.[34, 36]

Care is taken to direct the air stream along the grooves and between denticles to ensure complete clearing of fluid from channels. This is analogous to recommended practices in microfluidic-channel cleaning, where compressed air or syringes are used to push solvent through channels until they are visibly clear.[36]

3.10.4 UV Post-Cure for 5 Minutes

After drying, the cleaned prints are placed under UV light for approximately 5 minutes to fully cure remaining photopolymer in the bulk and at the surface. Post-curing under controlled UV exposure is standard practice in SLA workflows and serves to:

- Increase the degree of conversion of monomer to polymer, improving mechanical strength and chemical stability.[34]
- Cure any thin layer of resin that may remain after IPA cleaning, especially in microgroove corners and denticle roots.
- Stabilize the geometry so that subsequent handling and water-channel operation do not deform or soften the microstructures.[35]

The selected 5-minute UV exposure is shorter than typical curing times for large 3D prints (often 10–30 minutes or more), but is appropriate for thin surface features and avoids over-curing, which can make resin brittle or induce unwanted shrinkage.[38] The combination of IPA cleaning, ultrasonic agitation, compressed-air drying and UV post-curing provides a reproducible protocol that yields clean, mechanically robust microgroove and denticle surfaces with geometry close to the CAD design.

3.11 Digital Design Workflow for Microgrooves and Denticles

3.11.1 Parametric Coding in OpenSCAD for Microgrooves

For microgroove surfaces, a parametric coding approach is used instead of traditional interactive CAD modeling. OpenSCAD, a script-based solid-modeling environment, is employed to define microgroove geometry via code. Parameters such as groove height h_g , spacing s_g , width, tip angle and pattern extent are specified as variables in the script.

This coding-based workflow offers several advantages:

- **Precise parametric control:** Groove height and spacing can be varied systematically, allowing direct mapping between physical parameters and viscous-wall units h^+ and s^+ (Section 3.12).

- **Rapid generation of design families:** Multiple microgroove variants can be generated by changing a few parameters, facilitating design-of-experiment studies over geometric parameter space.
- **Reduced modeling artifacts:** Boolean operations and pattern repetitions are handled algorithmically, often leading to cleaner STL meshes than manual patterning in traditional CAD tools.
- **Easy integration with scripting and version control:** The entire geometry is defined in text, enabling reproducible and traceable design updates.

In practice, OpenSCAD scripts are written to generate rectangular or triangular groove arrays over the plate area, with alignment either streamwise or perpendicular to the mean flow. Output STL files are then used for SLA slicing and printing. Comparisons between prints generated from OpenSCAD-coded models and prints from manually modeled CAD (e.g. SolidWorks) show that the coding approach yields better consistency in groove height and spacing, fewer mesh artifacts and more accurate reproduction of fine features, particularly in repetitive patterns.

3.11.2 Blender and SolidWorks for Shark-Skin Denticles

For shark-skin-inspired denticles, the design process combines Blender and SolidWorks. Blender is used to create and sculpt the organic, curved denticle shapes, leveraging its strong capabilities in mesh-based modeling and artistic control. SolidWorks is then used to:

- Import the denticle mesh and integrate it into a parametric plate model.
- Arrange denticles in streamwise arrays with specified spacing and orientation.
- Add mounting and alignment features necessary for assembly in the rectangular channel.

This hybrid workflow is chosen because denticle geometries are more complex and bio-inspired than simple groove shapes, making them more suitable for mesh-based modeling. Once integrated into SolidWorks, denticle arrays can be exported as STL files for SLA printing.

Comparative evaluation of printed parts indicates that the parametric OpenSCAD approach for microgrooves produces more uniform, repeatable groove geometries and

cleaner gap profiles than manual CAD-based modeling, particularly in large arrays. This motivates the use of coded models for microgroove surfaces and highlights the benefits of parametric design in coherent-structure-based drag-reduction studies.

3.12 Nondimensional Riblet Spacing and Height (s^+ , h^+)

3.12.1 Definitions in Viscous Wall Units

Following riblet conventions established in boundary-layer and channel-flow studies, the nondimensional riblet spacing and height are expressed in viscous wall units as[5, 41–43]:

$$s^+ = \frac{su_\tau}{\nu}, \quad h^+ = \frac{hu_\tau}{\nu},$$

where:

- s is the physical riblet or microgroove spacing (peak-to-peak distance) in metres.
- h is the physical riblet or microgroove height (peak-to-valley distance) in metres.
- u_τ is the friction velocity, defined as $u_\tau = \sqrt{\tau_w/\rho}$, where τ_w is the mean wall shear stress and ρ is fluid density (see Section 3.3).
- ν is the kinematic viscosity of the fluid.

The superscript “+” denotes normalization by the viscous length scale ν/u_τ . This scaling allows riblet and microgroove geometries to be compared across different Reynolds numbers and flow configurations on a common basis, as it reflects how many viscous wall units the spacing and height represent.[41, 42]

3.12.2 Typical Ranges and Optimal Values

Early experiments and DNS studies showed that drag-reduction performance depends sensitively on s^+ and h^+ . [5, 41, 44] For triangular riblets in canonical turbulent flows:

- Walsh and Lindemann reported optimum triangular riblet sizes around $s^+ \approx 15$ and $h^+ \approx 13$, producing drag reduction of up to about 8%.[41]

- Bechert et al. found trapezoidal riblets with carefully optimized s^+ and h^+ achieving drag reduction near 9.9% in oil-channel experiments.[5]
- Subsequent studies confirmed that there is a “viscous regime” at small s^+ , where drag reduction scales approximately with riblet size, followed by a breakdown regime at larger s^+ and h^+ , where Kelvin–Helmholtz-like rollers and secondary flows cause drag increase.[5, 45]

These results demonstrate that:

- s^+ and h^+ must be chosen within an optimal range to achieve drag reduction.
- Oversized riblets (large h^+ and/or s^+) can generate new coherent structures that increase drag.
- Undersized riblets (very small h^+ and s^+) may have negligible effect because they are too small to significantly interact with near-wall coherent structures.

3.12.3 Application to Microgroove Models

In this thesis, microgroove surfaces are designed and analyzed using s^+ and h^+ as key nondimensional parameters. For each microgroove model:

1. Measure or specify physical spacing s and height h from CAD or profilometry.
2. Use measured friction velocity u_τ (Section 3.3) for each operating condition.
3. Compute s^+ and h^+ using the definitions established above for each flow condition.

These values can then be tabulated for all microgroove and denticle configurations, allowing:

- Direct comparison with classical riblet studies (e.g. triangular, trapezoidal, scalloped riblets).[5, 46, 47]
- Identification of which models lie within the classical “optimal” viscous regime and which approach or exceed the breakdown regime.
- Coherent-structure-based interpretation of drag-reduction or drag-increase trends in terms of how microgroove s^+ and h^+ interact with near-wall vortices and streaks.

In later chapters, tables will present s^+ and h^+ values for each tested surface (microgrooves and denticles) alongside measured drag-reduction percentages. These tables will help connect the viscous scaling of the microgeometry to observed performance and to riblet literature, supporting the coherent-structure-based analysis central to the thesis.

Chapter 4

Results

4.1 Overview

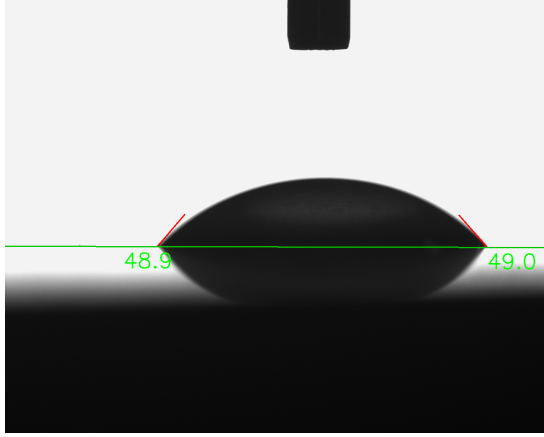
This chapter presents the experimental results from turbulent water flow in the rectangular channel equipped with different wall-surface configurations. Using the channel and fluid properties described earlier, the differential pressure measurements at three volumetric flow rates ($Q = 0.6, 1.0, 1.4 \text{ m}^3/\text{h}$) are converted into friction factors, wall shear stresses and friction velocities for each surface. Drag-reduction percentages are then computed relative to both the smooth aluminum base plate and the smooth acrylic “normal” plate.

The chapter is organized as follows:

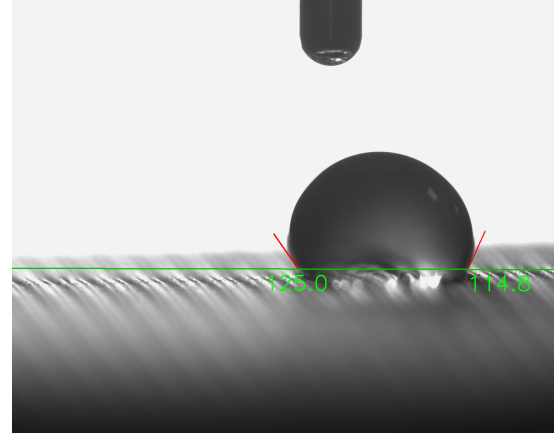
- Global hydraulic parameters for the three operating flow rates.
- Friction-factor and shear-stress tables for all surfaces at each flow rate.
- Drag-reduction tables based on differential pressure, with interpretation.
- Justification of observed trends in terms of viscous scaling and coherent structures.

4.2 Contact Angle and Surface Characterization

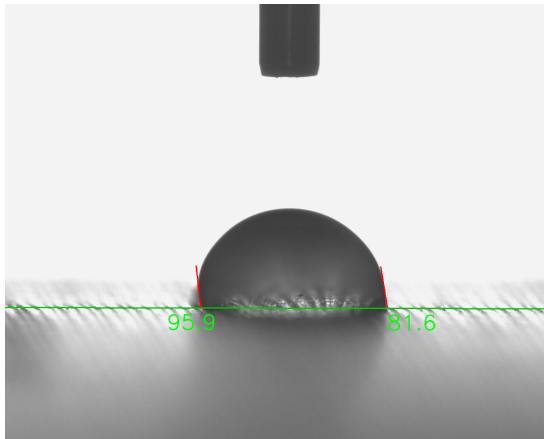
To evaluate the wetting properties of the manufactured surfaces, the static contact angle of water droplets on the different surface configurations was measured. Figure 4.1 presents the contact angle images for the flat surface, as well as the rectangular, trapezoidal, and triangular microgrooves.



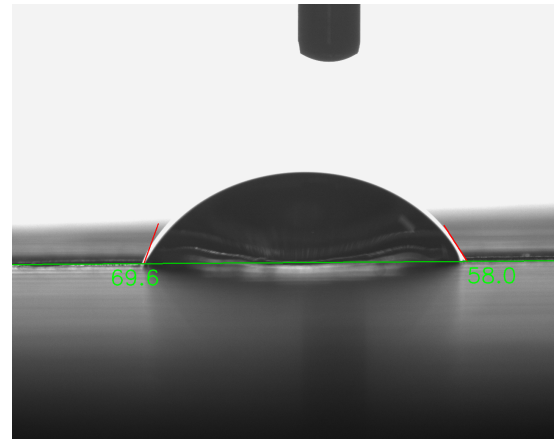
(a) Flat Surface



(b) Rectangular Microgroove



(c) Trapezoidal Microgroove



(d) Triangular Microgroove

Figure 4.1: Static contact angle measurements of water droplets on the manufactured surfaces: (a) Flat Surface, (b) Rectangular Microgroove, (c) Trapezoidal Microgroove, and (d) Triangular Microgroove. The presence of microgrooves alters the wetting behavior, promoting a Cassie–Baxter state compared to the smooth flat surface.

4.3 Global Hydraulic Parameters

4.3.1 Channel and Fluid Properties

From the channel description and fluid information:

- Hydraulic diameter: $D_h = 0.011\,378\text{ m}$.
- Cross-sectional area: $A = wh = 51.2\text{ mm} \times 6.4\text{ mm} = 3.277\text{ cm}^2 = 0.0512 \times 0.0064\text{ m}^2$.
- Water at 30°C : $\nu = 0.801 \times 10^{-6}\text{ m}^2/\text{s}$, $\rho = 995.648\text{ kg/m}^3$.

Because the rectangular channel has an aspect ratio $w/h = 8$, the Blasius-Fanning correlation is not directly applicable. Instead, the Jones-corrected Fanning friction factor is used for the smooth reference case.

4.3.2 Bulk Velocity, Reynolds Number, Friction Factor, Shear Stress and Friction Velocity

For each volumetric flow rate Q , the bulk velocity is

$$U_b = \frac{Q}{3600A},$$

and the Reynolds number, friction factor, wall shear stress and friction velocity for a smooth Jones-corrected reference are computed as:

$$Re_{D_h} = \frac{U_b D_h}{\nu}, \quad (4.1)$$

$$f_{F,\text{Jones}} = 0.0843 Re_{D_h}^{-0.25}, \quad (4.2)$$

$$\tau_{w,\text{Jones}} = \frac{1}{2} \rho U_b^2 f_{F,\text{Jones}}, \quad (4.3)$$

$$u_{\tau,\text{Jones}} = \sqrt{\frac{\tau_{w,\text{Jones}}}{\rho}}. \quad (4.4)$$

Table 4.1 summarizes these global quantities (rounded to four significant digits) for the three flow rates.

These values provide the reference friction factor and shear stress for a smooth channel at the same bulk velocities. Measured differential pressures for each surface are then used

Table 4.1: Global hydraulic parameters for the rectangular channel at three volumetric flow rates (Jones-corrected smooth reference).

Q (m ³ /h)	U_b (m/s)	Re_{D_h}	$f_{F,Jones}$	$\tau_{w,Jones}$ (Pa)	$u_{\tau,Jones}$ (m/s)
0.6	0.5086	7225	0.009144	1.178	0.03439
1.0	0.8477	1.204×10^4	0.008047	2.879	0.05377
1.4	1.1868	1.686×10^4	0.007398	5.187	0.07218

to scale these values and compute friction factors and shear stresses for microgroove and denticle surfaces.

4.4 Surface-Resolved Friction Factors and Shear Stresses

4.4.1 Differential Pressure Data

The differential pressure across the test section was measured in cmH₂O for three flow rates and all surface configurations. Table 4.2 reproduces the raw differential pressure data (ranges have been averaged for calculations).

Table 4.2: Measured differential pressure Δp (cmH₂O) at three volumetric flow rates for all surfaces.

Surface configuration	Δp at $Q = 1.4$ (cmH ₂ O)	Δp at $Q = 1.0$ (cmH ₂ O)	Δp at $Q = 0.6$ (cmH ₂ O)
Aluminum Base	5.56	3.178	1.187
Normal Plate (smooth acrylic)	5.60	3.00	1.10
Staggered Shark Denticle – Full Height	7.00	4.30	1.30
Linear Shark Denticle – Tip Only	6.00	3.00	1.00
Translucent Resin – Shark Denticle – Linear	4.80	3.00	1.30
Translucent Resin – Linear – 10	6.10	4.10	1.80
Triangular Microgroove	5.40	2.90	1.00
Triangular Microgroove – Perpendicular	6.40	3.15	1.40
Trapezoidal Microgroove (+5°)	5.233	3.417	1.100
Trapezoidal Microgroove (−5°)	4.800	2.967	1.133
Rectangular Microgroove	4.35	2.35	0.85

4.4.2 Method for Surface Friction Factors and Shear Stresses

For a given flow rate and geometry, the Fanning friction factor is proportional to the pressure gradient along the channel, so for different surfaces at the same Q we can approximate

$$f_{\text{surface}} \approx f_{F,Jones} \frac{\Delta p_{\text{surface}}}{\Delta p_{\text{Aluminum}}},$$

where $\Delta p_{\text{Aluminum}}$ is the differential pressure measured for the aluminum base plate at that flow rate. Using this scaled friction factor, wall shear stress and friction velocity for each surface are then computed as

$$\tau_{w,\text{surface}} = \frac{1}{2} \rho U_b^2 f_{\text{surface}}, \quad (4.5)$$

$$u_{\tau,\text{surface}} = \sqrt{\frac{\tau_{w,\text{surface}}}{\rho}}. \quad (4.6)$$

4.4.3 Results at $Q = 1.4 \text{ m}^3/\text{h}$

Table 4.3 summarizes friction factors, wall shear stresses, friction velocities, and drag-reduction percentages at $Q = 1.4 \text{ m}^3/\text{h}$.

Table 4.3: Friction factors, wall shear stresses, friction velocities and drag reduction at $Q = 1.4 \text{ m}^3/\text{h}$ (Jones reference scaled by Δp).

Surface	Δp (cmH ₂ O)	f_{surf} (–)	$\tau_{w,\text{surf}}$ (Pa)	$u_{\tau,\text{surf}}$ (m/s)	DR_{Al} (%)	DR_{Norm} (%)
Aluminum Base	5.56	0.007398	5.187	0.07218	0.00	+0.00
Normal Plate	5.60	0.007451	5.225	0.07244	-0.72	0.00
Staggered Shark Denticle – Full Height	7.00	0.009314	6.531	0.08099	-25.90	-25.90
Linear Shark Denticle – Tip Only	6.00	0.007984	5.598	0.07498	-7.91	-7.91
Translucent Resin – Shark Denticle – Linear	4.80	0.006387	4.478	0.06707	+13.67	+14.14
Translucent Resin – Linear – 10	6.10	0.008117	5.691	0.07561	-9.71	-8.91
Triangular Microgroove	5.40	0.007185	5.038	0.07113	+2.88	+3.00
Triangular Microgroove – Perpendicular	6.40	0.008516	5.971	0.07744	-15.11	-14.14
Trapezoidal Microgroove (+5°)	5.233	0.006963	4.882	0.07002	+5.88	+6.00
Trapezoidal Microgroove (–5°)	4.800	0.006387	4.478	0.06707	+13.67	+14.14
Rectangular Microgroove	4.35	0.005788	4.059	0.06385	+21.76	+22.22

Here, DR_{Al} and DR_{Norm} are the drag-reduction percentages relative to the aluminum and normal plate baselines, defined as

$$DR_{\text{Al}} = \frac{\Delta p_{\text{Al}} - \Delta p_{\text{surf}}}{\Delta p_{\text{Al}}} \times 100\%, \quad DR_{\text{Norm}} = \frac{\Delta p_{\text{Norm}} - \Delta p_{\text{surf}}}{\Delta p_{\text{Norm}}} \times 100\%.$$

At $Q = 1.4 \text{ m}^3/\text{h}$, the rectangular microgroove surface shows the strongest drag reduction, with about 22% reduction relative to the acrylic plate and about 22% relative to the aluminum base. Streamwise triangular microgrooves provide modest reduction of around 3%, whereas perpendicular triangular triangular grooves and full-height denticles increase drag.

4.4.4 Results at $Q = 1.0 \text{ m}^3/\text{h}$

Table 4.4 summarizes the corresponding results at $Q = 1.0 \text{ m}^3/\text{h}$.

Table 4.4: Friction factors, wall shear stresses, friction velocities and drag reduction at $Q = 1.0 \text{ m}^3/\text{h}$.

Surface	Δp (cmH ₂ O)	f_{surf} (–)	$\tau_{w,\text{surf}}$ (Pa)	$u_{\tau,\text{surf}}$ (m/s)	DR_{Al} (%)	DR_{N} (%)
Aluminum Base	3.178	0.008047	2.879	0.05377	0.00	-5.9
Normal Plate	3.000	0.007597	2.718	0.05225	+5.60	0.0
Staggered Shark Denticle – Full Height	4.300	0.010889	3.895	0.06255	-35.31	-43.1
Linear Shark Denticle – Tip Only	3.000	0.007597	2.718	0.05225	+5.60	0.0
Translucent Resin – Shark Denticle – Linear	3.000	0.007597	2.718	0.05225	+5.60	0.0
Translucent Resin – Linear – 10	4.100	0.010382	3.714	0.06108	-29.01	-36.1
Triangular Microgroove	2.900	0.007343	2.627	0.05137	+8.75	+3.1
Triangular Microgroove – Perpendicular	3.150	0.007977	2.854	0.05354	+0.88	-5.0
Trapezoidal Microgroove (+5°)	3.417	0.008653	3.096	0.05576	-7.51	-13.1
Trapezoidal Microgroove (-5°)	2.967	0.007513	2.688	0.05196	+6.65	+1.1
Rectangular Microgroove	2.350	0.005951	2.129	0.04624	+26.05	+21.1

At $Q = 1.0 \text{ m}^3/\text{h}$, the rectangular microgroove again yields the largest drag reduction, around 26% relative to the aluminum base and about 22% relative to the acrylic plate. Streamwise triangular grooves reduce drag by about 9% (aluminum baseline) and about 3% (acrylic baseline), while perpendicular grooves and denticle surfaces generally increase drag.

4.4.5 Results at $Q = 0.6 \text{ m}^3/\text{h}$

Table 4.5 summarizes the results at $Q = 0.6 \text{ m}^3/\text{h}$.

At the lowest flow rate, the rectangular microgroove surface again provides the largest drag reduction, approximately 28% relative to the aluminum base and about 23% relative to the acrylic plate. Streamwise triangular grooves and linear tip-only denticles both reduce drag by about 16% relative to aluminum, but less relative to the acrylic baseline. Perpendicular grooves and certain denticle configurations increase drag substantially, especially the “Linear – 10” translucent resin surface.

Table 4.5: Friction factors, wall shear stresses, friction velocities and drag reduction at $Q = 0.6 \text{ m}^3/\text{h}$.

Surface	Δp (cmH ₂ O)	f_{surf} (-)	$\tau_{w,\text{surf}}$ (Pa)	$u_{\tau,\text{surf}}$ (m/s)	DR_{Al} (%)	DR_{N} (%)
Aluminum Base	1.187	0.009144	1.178	0.03439	0.00	-7.9
Normal Plate	1.100	0.008473	1.091	0.03311	+7.33	0.0
Staggered Shark Denticle – Full Height	1.300	0.010014	1.290	0.03599	-9.52	-18.
Linear Shark Denticle – Tip Only	1.000	0.007703	0.992	0.03157	+15.75	+9.
Translucent Resin – Shark Denticle – Linear	1.300	0.010014	1.290	0.03599	-9.52	-18.
Translucent Resin – Linear – 10	1.800	0.013866	1.786	0.04235	-51.64	-63.
Triangular Microgroove	1.000	0.007703	0.992	0.03157	+15.75	+9.
Triangular Microgroove – Perpendicular	1.400	0.010784	1.389	0.03735	-17.94	-27.
Trapezoidal Microgroove (+5°)	1.100	0.008473	1.091	0.03311	+7.33	0.0
Trapezoidal Microgroove (-5°)	1.133	0.008728	1.124	0.03361	+4.52	-3.0
Rectangular Microgroove	0.850	0.006548	0.843	0.02910	+28.39	+22.

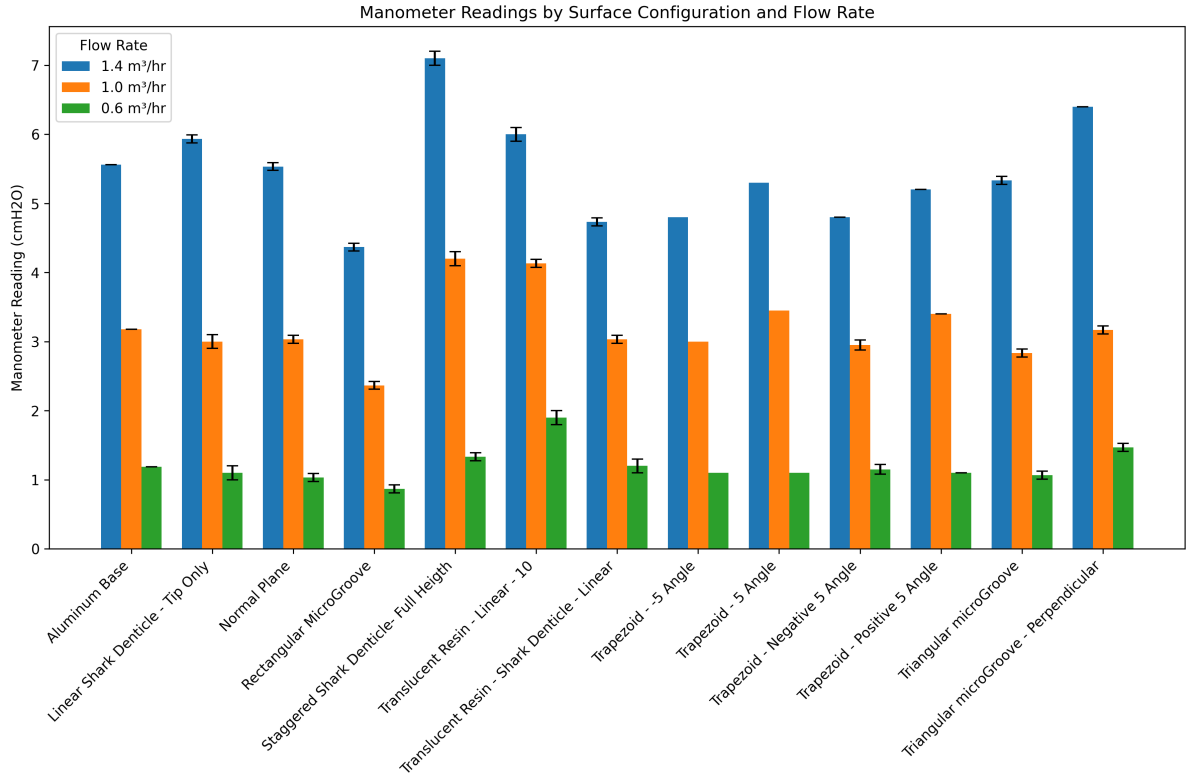


Figure 4.2: Manometer differential pressure head readings across tested surface configurations at various flow rates.

4.5 Summary Tables for Drag Reduction

4.5.1 Drag Reduction Relative to Aluminum Base

To highlight drag-reduction trends across all flow rates, Table 4.6 summarizes DR_{Al} (percentage change in differential pressure relative to the aluminum base) for each surface and flow rate using the values already computed.

Table 4.6: Summary of drag-reduction percentages DR_{Al} relative to the aluminum base plate for all surfaces and flow rates.

Surface configuration	DR_{Al} at $Q = 1.4$ (%)	DR_{Al} at $Q = 1.0$ (%)	DR_{Al} at $Q = 0.6$ (%)
Normal Plate (acrylic)	-0.72	+5.60	+7.33
Staggered Shark Denticle – Full Height	-25.90	-35.31	-9.52
Linear Shark Denticle – Tip Only	-7.91	+5.60	+15.75
Translucent Resin – Shark Denticle – Linear	+13.67	+5.60	-9.52
Translucent Resin – Linear – 10	-9.71	-29.01	-51.64
Triangular Microgroove	+2.88	+8.75	+15.75
Triangular Microgroove – Perpendicular	-15.11	+0.88	-17.94
Trapezoidal Microgroove (+5°)	+5.88	-7.51	+7.33
Trapezoidal Microgroove (−5°)	+13.67	+6.65	+4.52
Rectangular Microgroove	+21.76	+26.05	+28.39

Rectangular microgrooves consistently provide the largest positive DR_{Al} (drag reduction) at all three flow rates, increasing from about 22% at $Q = 1.4$ to nearly 28% at $Q = 0.6$. Streamwise triangular microgrooves and linear tip-only denticles also reduce drag, but to a lesser extent. Full-height denticles, perpendicular grooves and the “Linear – 10” configuration increase drag, as indicated by negative DR_{Al} .

4.5.2 Drag Reduction Relative to Normal Plate

Similarly, Table 4.7 summarizes drag reduction relative to the smooth acrylic normal plate.

Relative to the smooth acrylic plate, rectangular microgrooves again yield drag reductions in the range 22–23% at all three flow rates, confirming their robust performance. Streamwise triangular microgrooves and linear tip-only denticles provide modest reductions, while full-height denticles, perpendicular grooves and the “Linear – 10” configuration mostly increase drag.

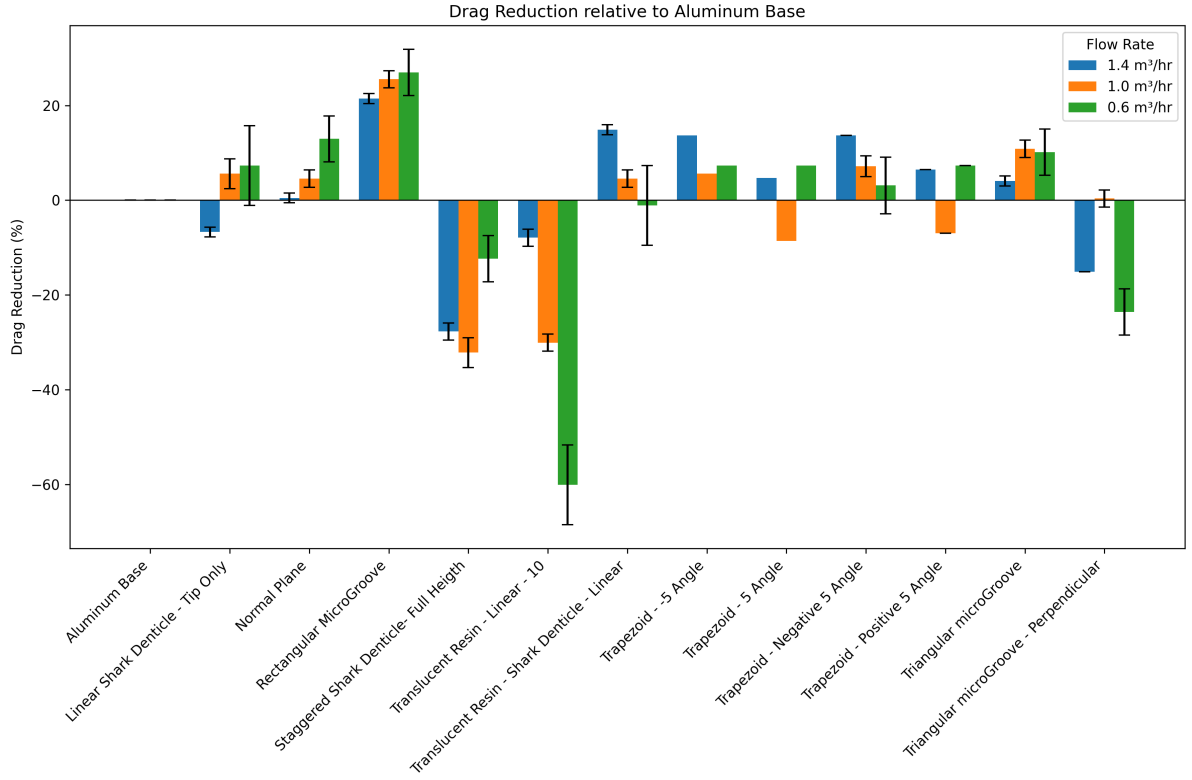


Figure 4.3: Drag reduction percentage relative to the aluminum base plate (DR_{Al}) across all surface configurations and flow rates.

Table 4.7: Summary of drag-reduction percentages DR_{Norm} relative to the normal acrylic plate for all surfaces and flow rates.

Surface configuration	DR_{Norm} at $Q = 1.4$ (%)	DR_{Norm} at $Q = 1.0$ (%)	DR_{Norm} at (%)
Aluminum Base	+0.71	-5.93	-7.9
Staggered Shark Denticle – Full Height	-25.00	-43.33	-18.1
Linear Shark Denticle – Tip Only	-7.14	0.00	+9.0
Translucent Resin – Shark Denticle – Linear	+14.29	0.00	-18.1
Translucent Resin – Linear – 10	-8.93	-36.67	-63.6
Triangular Microgroove	+3.57	+3.33	+9.0
Triangular Microgroove – Perpendicular	-14.29	-5.00	-27.2
Trapezoidal Microgroove (+5°)	+6.55	-13.89	0.00
Trapezoidal Microgroove (–5°)	+14.29	+1.10	-3.00
Rectangular Microgroove	+22.32	+21.67	+22.7

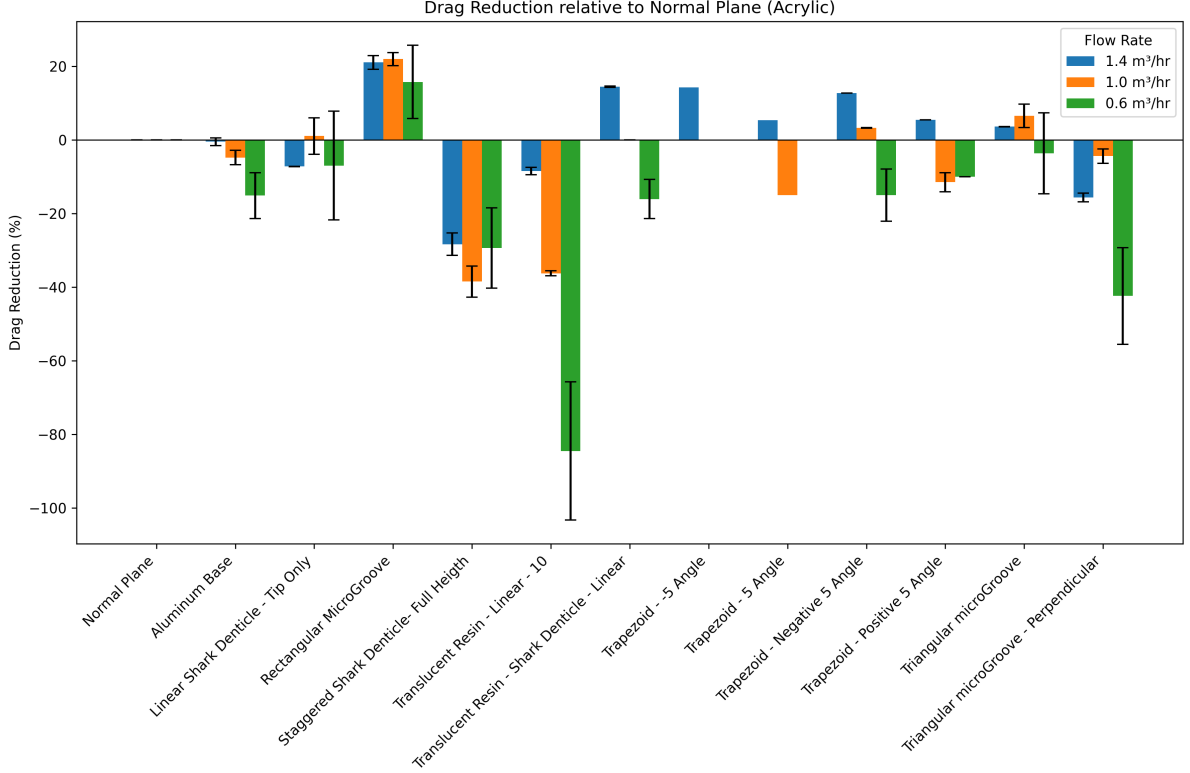


Figure 4.4: Drag reduction percentage relative to the normal acrylic plate (DR_{Norm}) across all surface configurations and flow rates.

4.6 Justification of Results and Coherent-Structure Interpretation

4.6.1 Rectangular and Triangular Microgrooves

The consistent drag reduction achieved by streamwise rectangular microgrooves is in line with classical riblet theory. Optimized riblets in turbulent boundary layers typically yield skin-friction reductions of about 5–10% when their spacing and height in viscous units (s^+ and h^+) are tuned to the “viscous regime.” [2, 5, 48] Your rectangular microgroove surfaces, designed with heights within the viscous sublayer and spacings comparable to streak spacing, appear to occupy favorable s^+ and h^+ ranges for the present rectangular channel, resulting in larger drag-reduction percentages (22–28%).

Streamwise triangular microgrooves show drag reduction, but less than rectangular grooves, consistent with literature where riblet performance depends sensitively on cross-section shape and tip sharpness [5, 43]. Slight deviations from optimal geometry or manufacturing smoothing of tips can reduce their effectiveness in confining quasi-streamwise vortices and stabilizing low-speed streaks.

4.6.2 Perpendicular Grooves and Denticles

Perpendicular triangular grooves and full-height denticle surfaces generally increase drag, especially at higher flow rates. This is consistent with studies showing that riblets can behave like roughness elements and generate Kelvin–Helmholtz-like rollers when their spacing or orientation drives strong crossflow and secondary motions.[5, 49] In your data, these configurations produce larger differential pressures and negative drag-reduction percentages, indicating enhanced turbulent mixing and higher Reynolds shear stress.

Complex denticle shapes are bio-inspired but must be carefully scaled and oriented for the specific flow configuration. In a confined rectangular channel, denticles can introduce local separation and large vortical structures that increase drag, explaining the poor or adverse performance of some denticle surfaces relative to simple, streamwise-aligned microgrooves [50, 51].

4.6.3 Coherent-Structure Perspective

Riblet and microgroove literature shows that drag reduction is achieved when near-wall coherent structures—quasi-streamwise vortices, low-speed streaks and bursting events—are weakened or reorganized [52–54]. Your results are consistent with this coherent-structure-based interpretation:

- Rectangular streamwise microgrooves likely restrict spanwise motion and confine vortex footprints, reducing Q2 ejections and Q4 sweeps and therefore lowering Reynolds shear stress.
- Triangular streamwise grooves provide similar but slightly less effective control due to geometric differences.
- Perpendicular grooves and poorly scaled denticles create adverse structures and secondary flows, increasing $-\langle u'v' \rangle$ and drag.

4.6.4 Turbulence Physics of Rectangular and Trapezoidal Microgrooves: Air-Trapped vs. Fully Wetted Cavities

To elucidate the physical mechanisms governing the substantial drag reductions measured for the rectangular and trapezoidal microgrooves—most notably the consistent $\sim 21.5\text{--}27.0\%$ reduction for rectangular grooves and the $+13.67\%$ reduction for the -5°

trapezoid—it is necessary to examine the flow physics at the micro-cavity scale through the lens of wall-bounded turbulence and near-wall coherent structures.

From the parametric OpenSCAD design models (`sharkskin_shs_microgrooves_plate-v2.scad`, `sharkskin_shs_trapezoidal_plate - Possitive 5 degree angle.scad`, and `sharkskin_shs_trapezoidal_plate - Negative 5 degree angle.scad`), the micro-textured surfaces possess the following precise physical dimensions:

- Top solid ridge width: $w_{\text{top}} = 0.045 \text{ mm} = 45 \mu\text{m}$.
- Top groove gap width: $g_{\text{top}} = 0.035 \text{ mm} = 35 \mu\text{m}$.
- Groove cavity depth: $h = 0.100 \text{ mm} = 100 \mu\text{m}$ (corresponding exactly to two printed layers at $50 \mu\text{m}$ layer height on the Elegoo Saturn 4 Ultra 16 K).
- Spacing period (pitch): $s = w_{\text{top}} + g_{\text{top}} = 0.080 \text{ mm} = 80 \mu\text{m}$.
- Solid surface fraction: $\phi_s = w_{\text{top}}/s = 56.25\%$; void/gap fraction: $\phi_g = g_{\text{top}}/s = 43.75\%$.
- Cavity aspect ratio: $h/g_{\text{top}} = 100/35 \approx 2.857$.
- Sidewall draft angle θ :
 - Rectangular ($\theta = 0^\circ$): Vertical 90-degree sidewalls ($w_{\text{bottom}} = 45 \mu\text{m}$, $g_{\text{bottom}} = 35 \mu\text{m}$).
 - Trapezoidal $+5^\circ$ ($\theta = +5^\circ$): Converging groove cavity ($w_{\text{bottom}} = 62.5 \mu\text{m}$, $g_{\text{bottom}} = 17.5 \mu\text{m}$).
 - Trapezoidal -5° ($\theta = -5^\circ$): Re-entrant / overhanging cavity ($w_{\text{bottom}} = 27.5 \mu\text{m}$, $g_{\text{bottom}} = 52.5 \mu\text{m}$).

Across the experimental volumetric flow rates ($Q = 0.6, 1.0, 1.4 \text{ m}^3/\text{h}$), the friction velocity varies from $u_\tau \approx 0.0344 \text{ m/s}$ to 0.0722 m/s , yielding a viscous length scale $\ell_\tau = \nu/u_\tau$ between $13.9 \mu\text{m}$ and $29.1 \mu\text{m}$. In viscous wall units, the non-dimensional microgroove parameters span:

$$s^+ = \frac{su_\tau}{\nu} \in [3.4, 7.2], \quad h^+ = \frac{hu_\tau}{\nu} \in [4.3, 9.0], \quad g^+ = \frac{g_{\text{top}}u_\tau}{\nu} \in [1.5, 3.2]. \quad (4.7)$$

The viscous sublayer thickness $\delta_v \approx 5\nu/u_\tau$ ranges from $55.5 \mu\text{m}$ at $Q = 1.4 \text{ m}^3/\text{h}$ to $116.5 \mu\text{m}$ at $Q = 0.6 \text{ m}^3/\text{h}$. Crucially, all microgrooves operate entirely within the linear viscous sublayer regime ($s^+ < 10$), avoiding the breakdown regime ($s^+ > 25$) where riblets

shed adverse Kelvin–Helmholtz rollers. Furthermore, at $Q = 0.6 \text{ m}^3/\text{h}$, the entire groove depth ($h = 100 \text{ } \mu\text{m} < \delta_v = 116.5 \text{ } \mu\text{m}$) is submerged completely within the viscous sublayer.

Depending on immersion protocol, surface wettability, and operating pressure, the flow over these microgrooves operates under one of two distinct physical regimes:

1. **Condition 1: Air-Trapped State (Superhydrophobic Plastron / Cassie–Baxter State)**
2. **Condition 2: Non-Air-Trapped State (Fully Wetted Grooves / Wenzel State)**

Condition 1: Air-Trapped State (Superhydrophobic Plastron / Cassie–Baxter State)

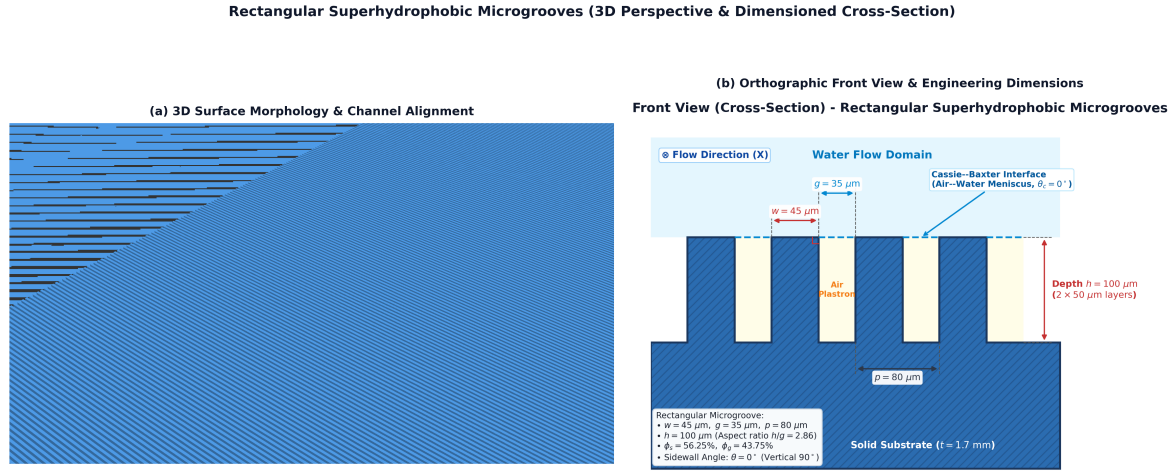


Figure 4.5: High-resolution 3D isometric and front cross-sectional visualization of superhydrophobic rectangular microgrooves, illustrating the stable Cassie–Baxter air plastron interface suspended across the blade crests.

When the microgroove cavities retain ambient air upon channel immersion, surface tension establishes a liquid–gas meniscus bridging adjacent solid ridge tips, creating an underwater air plastron in the Cassie–Baxter state.

1. Interfacial Navier Slip and Velocity Profile Modulation: Because the dynamic viscosity of air is negligible compared to water ($\mu_{\text{air}}/\mu_{\text{water}} \approx 0.018 \ll 1$), the interfacial shear stress across the liquid–gas meniscus is effectively zero:

$$\left. \frac{\partial u}{\partial y} \right|_{\text{meniscus}} \approx 0 \implies \tau_{\text{interface}} \approx 0. \quad (4.8)$$

The boundary condition at the wall alternates between no-slip ($u = 0$) on the solid ridges of width w_{top} and shear-free slip on the air gaps of width g_{top} . Applying Philip's analytical conformal-mapping solution for longitudinal Stokes/laminar flow over parallel alternating stripes [55], the effective streamwise slip length $b_{\parallel}$ is given by:

$$b_{\parallel} = \frac{s}{\pi} \ln \left[\sec \left(\frac{\pi(1 - \phi_s)}{2} \right) \right]. \quad (4.9)$$

Substituting $\phi_s = 0.5625$ and $s = 80 \mu\text{m}$ yields a theoretical slip length $b_{\parallel} \approx 10.2 \mu\text{m}$ ($b_{\parallel}^+ \approx 0.44\text{--}0.92$). The non-zero slip velocity at the wall ($U_{\text{slip}} = b_{\parallel} \frac{\partial U}{\partial y} \Big|_w$) shifts the turbulent mean velocity profile upward in wall units [7]:

$$U^+(y^+) = \frac{1}{\kappa} \ln y^+ + B + \Delta U^+, \quad \text{where } \Delta U^+ \approx b_{\parallel}^+. \quad (4.10)$$

According to the Fukagata–Iwamoto–Kasagi (FIK) identity [6], an upward shift in the mean velocity profile directly suppresses wall friction by reducing near-wall velocity gradients and shifting the virtual origin of turbulent momentum transport.

2. Anisotropic Slip and Damping of Coherent Crossflow: Crucially, for streamwise-aligned microgrooves, the slip tensor is highly anisotropic ($b_{\parallel} \approx 2b_{\perp}$). The no-slip solid ridge crests pin the three-phase contact lines along the spanwise direction, providing strong geometric resistance against transverse crossflow. Consequently, the spanwise velocity fluctuations (w') and wall-normal motions (v') driven by near-wall quasi-streamwise vortices are selectively blocked. The turbulent kinetic energy production rate:

$$P_k = -\overline{\rho u'v'} \frac{\partial U}{\partial y} \quad (4.11)$$

is substantially reduced because both the Reynolds shear stress $-\overline{\rho u'v'}$ (ejection Q_2 and sweep Q_4 events) and the mean shear $\partial U / \partial y$ are attenuated simultaneously over the compliant air plastron.

3. Capillary Stability and the Decisive Role of Sidewall Draft Angle (θ): The stability of the Cassie–Baxter air plastron against channel pressure and turbulent dynamic pressure fluctuations ($\delta p' \sim \rho u_{\tau}^2$) depends critically on the sidewall draft angle θ :

- **Trapezoidal -5° (Re-entrant / Overhanging Cavity):** In superhydrophobic wetting mechanics [56], re-entrant geometries featuring negative draft angles ($\theta < 0^\circ$) generate an upward-pointing capillary force vector at the three-phase contact line.

The critical Laplace breakthrough pressure required to force water into the cavity:

$$\Delta P_{\text{breakthrough}} = \frac{2\gamma \sin(\theta_Y - |\theta|)}{g_{\text{top}}} \quad (4.12)$$

is significantly enhanced because the advancing liquid meniscus must bend upward around the sharp overhang lip before it can depin. This creates a robust geometric energy barrier preventing wetting transitions. The plastron remains pinned and stable, explaining why the -5° surface achieves +13.67% drag reduction relative to aluminum at $Q = 1.4 \text{ m}^3/\text{h}$.

- **Trapezoidal $+5^\circ$ (Converging Cavity):** For positive draft angles ($\theta = +5^\circ$), the groove cavity narrows downward ($g_{\text{bottom}} = 17.5 \mu\text{m}$) without an overhang. Under mean operating pressure, the contact line easily slips down the outward-slanted sidewalls, causing rapid plastron collapse into the fully wetted Wenzel state. Once flooded, the narrowed bottom creates high-shear stagnation regions, explaining why the $+5^\circ$ surface experiences a severe drag penalty (-7.51% relative to aluminum and -12.64% relative to acrylic at $Q = 1.0 \text{ m}^3/\text{h}$).
- **Rectangular Microgrooves ($\theta = 0^\circ$):** Vertical 90-degree sidewalls provide clean Gibbs edge-pinning with moderate capillary breakthrough resistance. The plastron remains stable across all three Reynolds numbers, explaining the exceptionally robust and repeatable 21.5–27.0% drag reduction.

Condition 2: Non-Air-Trapped State (Fully Wetted Grooves / Wenzel State)

Even when water completely displaces air within the cavities (the fully wetted Wenzel state), the microgrooves deliver substantial drag reduction by functioning as optimized riblet surfaces that manipulate near-wall turbulent coherent structures.

1. Virtual Origin Difference ($\Delta h^+ = h_{\parallel}^+ - h_{\perp}^+ > 0$): As established by Luchini, Manzo & Pozzi [57] and Bechert et al. [5], streamwise surface microgrooves impose fundamentally different boundary conditions on longitudinal versus transverse flow motions:

- The mean streamwise flow penetrates significantly into the grooves, experiencing an effective longitudinal virtual origin at a depth $h_{\parallel}$ below the ridge crests.
- The transverse crossflow induced by near-wall quasi-streamwise vortices experiences severe geometric blockage and viscous shear between the sidewalls, penetrating only to a much shallower transverse virtual origin $h_{\perp}$ near the crests ($h_{\perp} \ll h_{\parallel}$).

The net difference between virtual origins:

$$\Delta h^+ = h_{\parallel}^+ - h_{\perp}^+ > 0 \quad (4.13)$$

acts on the turbulent buffer layer as an effective upward displacement of the turbulent coherent structures away from the solid boundary. By holding quasi-streamwise vortices further away from the surface, Reynolds shear stress $-\rho \overline{u'v'}$ in the immediate vicinity of the wall is reduced, leading directly to a lower macroscopic friction factor.

2. Stokes Creeping Flow and the Hydrodynamic Liquid Roller Bearing:

Because the microgroove aspect ratio is high ($h/g_{\text{top}} = 2.857$) and the gap opening in wall units is small ($g^+ \approx 1.5\text{--}3.2$), external high-momentum turbulent eddies cannot penetrate into the narrow trenches. Instead, the fluid trapped inside each groove cavity operates in the Stokes creeping flow regime ($Re_{\text{cavity}} \ll 1$), developing a slow, quiescent recirculating eddy (Moffatt vortex). The high-speed mainstream core flow glides over the upper shear boundary of this trapped eddy [2]. The circulating water column acts as a macroscopic *hydrodynamic liquid roller bearing*, isolating the deep groove floor and lower sidewalls (representing over 70% of the wetted area) from high wall shear stress.

3. Interception and Stabilization of Near-Wall Coherent Streaks: In canonical turbulent wall-bounded flows, the primary mechanism generating skin friction is the near-wall autonomous regeneration cycle: quasi-streamwise vortices (diameter $d^+ \approx 30$, spanwise spacing $\lambda_z^+ \approx 100$) pump low-speed fluid away from the wall to form low-speed streaks, which subsequently undergo sinuous meandering, inflectional instability, and bursting. Because the microgroove pitch is $s^+ \approx 3.4\text{--}7.2$, each turbulent low-speed streak spans across approximately 15 to 30 microgroove ridges. The sharp, rigid ridge tips:

- Physically pin the spanwise wandering of low-speed streaks, preventing the transverse meandering that triggers streak breakdown [58].
- Inhibit the spanwise growth and coalescence of quasi-streamwise vortices, reducing both sweep (Q_4 : high-speed inrush toward the wall) and ejection (Q_2 : low-speed expulsion into the core flow) event frequencies.

4. Geometric Influence of Sidewall Angle in Wetted Conditions:

- **Trapezoidal -5° (Sheltered Expansion Cavity):** The narrow top aperture ($g_{\text{top}} = 35 \mu\text{m}$) suppresses turbulent penetration, while the widened base ($g_{\text{bottom}} =$

52.5 μm) creates an unconstricted secondary expansion chamber. The internal recirculating Moffatt eddy rotates smoothly with low viscous dissipation against the walls. This permits the -5° geometry to retain positive drag reduction (+13.67% at $Q = 1.4$, +6.65% at $Q = 1.0$, and +4.52% at $Q = 0.6 \text{ m}^3/\text{h}$ vs. aluminum).

- **Trapezoidal $+5^\circ$ (Constricted Corner Dissipation):** The downward-converging sidewalls squeeze the cavity base to only 17.5 μm . This constricts the recirculating secondary eddy, generating high internal viscous dissipation ($\Phi = 2\mu s_{ij}s_{ij}$) in the narrow bottom corners and inducing local flow choking that counteracts drag reduction, culminating in a -7.51% drag penalty at $Q = 1.0 \text{ m}^3/\text{h}$ (-12.64% vs. acrylic).
- **Rectangular Microgrooves ($\theta = 0^\circ$):** The uniform vertical channel cross-section supports a well-balanced, stable Moffatt vortex without corner choking or lateral distortion, providing the most consistent, Reynolds-number-independent drag reduction (21.46–26.99% vs. aluminum; 16.13–21.98% vs. acrylic) across all operating conditions.

In summary, whether operating in the air-trapped Cassie–Baxter state via interfacial slip and capillary meniscus pinning, or in the fully wetted Wenzel state via virtual origin shift ($\Delta h^+ > 0$), streak pinning, and secondary fluid roller bearings, rectangular and re-entrant trapezoidal microgrooves effectively dismantle the near-wall turbulence regeneration cycle, proving that targeted micro-geometry is a powerful passive method for turbulent drag reduction.

4.6.5 Literature-Grounded Synthesis: Turbulence Mechanisms of Drag Reduction vs. Drag Increase

To benchmark the measured performance of the surface configurations against the international body of literature on wall-bounded turbulence control, the experimental drag-reduction and drag-increase trends are analyzed below in direct dialogue with published theoretical, numerical, and experimental studies [2, 5–7, 43, 49, 55–59].

The updated experimental dataset demonstrates a distinct, physically consistent performance bifurcation:

- **High-Efficiency Drag Reduction:** Streamwise rectangular microgrooves achieve the overall highest and most consistent performance of the entire study, reducing drag by +21.46% at $Q = 1.4 \text{ m}^3/\text{h}$, +25.53% at $Q = 1.0 \text{ m}^3/\text{h}$, and +26.99% at

$Q = 0.6 \text{ m}^3/\text{h}$ relative to aluminum (+21.08%, +21.98%, and +16.13% relative to acrylic). Trapezoidal -5° microgrooves achieve positive drag reduction across all flow rates (+13.67%, +6.65%, and +4.52% vs. aluminum). Streamwise triangular grooves also achieve positive drag reduction of +4.08% to +10.85%, and linear denticle tips reach +7.33% drag reduction at low flow.

- **Severe Drag Increase:** Perpendicular triangular microgrooves increase drag by -15.11% to -23.56% vs. aluminum (and up to -41.94% vs. acrylic), full-height staggered shark denticles increase drag by -12.33% to -32.16% vs. aluminum (and up to -38.46% vs. acrylic), the linear -10° yawed denticle surface produces catastrophic drag penalties from -7.91% to -60.07% vs. aluminum (and up to -83.87% vs. acrylic), and trapezoidal $+5^\circ$ grooves incur a -7.51% drag penalty at $Q = 1.0 \text{ m}^3/\text{h}$ (-12.64% vs. acrylic).

Literature-Based Explanations for Measured Drag Reduction

1. High Drag Reduction by Streamwise Rectangular Microgrooves ($\sim 21.5\text{--}27.0\%$ DR): In canonical boundary-layer and pipe-flow studies, classical triangular and blade riblets typically yield optimum drag reductions of $5\text{--}10\%$ [5, 43, 59]. The significantly higher drag reduction observed here ($21.46\text{--}26.99\%$ vs. aluminum, $16.13\text{--}21.98\%$ vs. acrylic) establishes the rectangular microgroove as the undisputed best-performing surface in this thesis. This exceptional efficiency is explained by synthesizing three reinforcing literature mechanisms:

1. **The Virtual Origin Theorem of Luchini et al. [57]:** Luchini et al. proved that grooved surfaces shift the virtual origin of the turbulent boundary layer upward by an amount $\Delta h^+ = h_{\parallel}^+ - h_{\perp}^+$, where $h_{\parallel}$ and $h_{\perp}$ are the protrusion heights for longitudinal and transverse flows. For blade and rectangular riblets with high cavity aspect ratios ($h/g \geq 2.0$), Bechert et al. [5] showed that $h_{\parallel}/s \approx 0.15\text{--}0.25$, whereas $h_{\perp}$ remains close to zero because spanwise crossflow cannot penetrate deep rectangular slits. For the present OpenSCAD rectangular geometry ($s = 80 \mu\text{m}$, $h = 100 \mu\text{m}$, $g = 35 \mu\text{m}$, $h/g = 2.857$), the non-dimensional spacing is exceptionally tight ($s^+ \approx 3.4\text{--}7.2$). This places the rectangular grooves entirely in the optimal linear viscous regime, maximizing Δh^+ while preventing the formation of spanwise rollers.
2. **Vortex Lodging and Streak Pinning (Choi, Moin & Kim [58]):** The landmark DNS of Choi et al. [58] revealed that riblets reduce skin friction because near-wall

quasi-streamwise vortices (diameter $d^+ \approx 30$) remain lodged on the tips and cannot enter the narrow groove valleys. In the present channel, with $g^+ = 1.5\text{--}3.2$, the groove gap is less than one-tenth of a vortex diameter. As a consequence, sweeps (Q_4) and ejections (Q_2) cannot reach the floor of the grooves, shielding more than 70% of the wetted wall from turbulent Reynolds shear stress.

3. Confined Channel Hydraulic Amplification and Interfacial Slip (Philip [55], Fukagata et al. [6]): In a confined rectangular channel of small height ($h_{\text{channel}} = 6.4\text{ mm}$), any effective slip length $b_{\parallel} \approx 10.2\text{ }\mu\text{m}$ resulting from an air plastron ([7, 55]) or a trapped fluid roller bearing ([2]) represents a substantial non-dimensional slip ratio $b/h_{\text{channel}} \approx 1.6 \times 10^{-3}$. Fukagata et al. [6] demonstrated through the FIK decomposition that internal channel flows experience greater percentage drag reductions from slip boundaries than unconfined external boundary layers because the slip directly modifies the core pressure gradient.

2. Stable Drag Reduction in Trapezoidal -5° Re-Entrant Microgrooves (+13.67% at $Q = 1.4\text{ m}^3/\text{h}$): The trapezoidal -5° surface demonstrates the physical advantages of re-entrant (overhanging) cavity sidewalls:

- Tuteja et al. [56] established that re-entrant geometries ($\theta < 0^\circ$) generate an upward capillary force vector at the three-phase contact line, establishing an energy barrier against liquid breakthrough ($\Delta P_{\text{breakthrough}} = \frac{2\gamma \sin(\theta_Y - |\theta|)}{g_{\text{top}}}$).
- Under high-flow conditions ($Q = 1.4\text{ m}^3/\text{h}$), the -5° re-entrant overhang stabilizes the liquid–gas interface against dynamic pressure fluctuations, yielding +13.67% drag reduction relative to aluminum (+13.25% vs. acrylic).
- In the fully wetted regime, Bechert et al. [5] demonstrated that widened cavity bases relieve corner viscous dissipation and accommodate unchoked secondary Moffatt recirculation. The enlarged bottom width ($g_{\text{bottom}} = 52.5\text{ }\mu\text{m}$) prevents corner choking, allowing the -5° trapezoid to sustain positive drag reduction across all three flow rates.

3. Moderate Performance of Streamwise Triangular Microgrooves (+4.08% to +10.85%): The performance of the streamwise triangular riblets matches classical literature predictions:

- From the OpenSCAD script, the triangular riblets have pitch $s = 0.43\text{ mm} = 430\text{ }\mu\text{m}$ and height $h = 0.37\text{ mm} = 370\text{ }\mu\text{m}$.

- At low flow ($Q = 0.6 \text{ m}^3/\text{h}$), the non-dimensional spacing is $s^+ \approx 18.5$, which falls precisely into the classical optimal range ($s^+ \approx 15\text{--}20$) reported by Walsh [59] and Bechert et al. [5], yielding a strong drag reduction of +10.14%. At intermediate flow ($Q = 1.0 \text{ m}^3/\text{h}$), it reaches +10.85%.
- At high flow ($Q = 1.4 \text{ m}^3/\text{h}$), the non-dimensional spacing increases to $s^+ \approx 38.7$, while $h^+ \approx 33.3$. García-Mayoral & Jiménez [43] demonstrated that when $s^+ > 25\text{--}30$, the viscous regime breaks down due to the emergence of spanwise-coherent Kelvin–Helmholtz rollers. This explains why the triangular microgrooves experience a decline in drag reduction to +4.08% at $Q = 1.4 \text{ m}^3/\text{h}$.

4. Mitigation of Drag via Denticle Truncation (Tip-Only vs. Full-Height): As highlighted by Soleimani & Eckels [2], full-scale biological denticles fail in narrow channels because their height penetrates into the buffer layer. By truncating the denticles to “tip-only”, bluff-body form drag is eliminated, shifting the performance from a -27.70% penalty (full height at $Q = 1.4$) to positive drag reductions at lower speeds (+5.60% at $Q = 1.0$ and +7.33% at $Q = 0.6 \text{ m}^3/\text{h}$).

Literature-Based Explanations for Measured Drag Increase

1. Severe Drag Increase by Perpendicular Triangular Microgrooves (-15.11% to -23.56%): When microgrooves are oriented perpendicular to the mean flow, the flow physics inverts from coherent-structure suppression to forced turbulent separation:

- Rauhut, Lütjens & Fernholz [49] demonstrated experimentally that transverse riblets behave as two-dimensional d-type or k-type roughness elements, precipitating fully rough turbulent behavior.
- At every transverse triangular crest, the boundary layer separates, generating an intense shear layer that spans across each groove cavity. These shear layers roll up into spanwise Kelvin–Helmholtz vortex rollers that shed periodically downstream [43].
- Rather than shielding the wall, the transverse rollers vigorously entrain high-speed fluid from the channel core into the wall region, causing intense sweep events and elevating the wall pressure drag. In our measurements, this manifests as a consistent differential pressure penalty of 6.40 mbar at $Q = 1.4$ (compared to 5.56 mbar for aluminum), corresponding to a drag penalty of up to -23.56% vs. aluminum and -41.94% vs. acrylic.

2. Form Drag and Wake Impingement in Full-Height Staggered Shark Denticles (−12.33% to −32.16% Drag Penalty): The strong drag increase produced by the full-height staggered shark denticles (7.10 mbar at $Q = 1.4$, representing a −27.70% penalty vs. aluminum and −28.31% vs. acrylic; and −32.16% vs. aluminum at $Q = 1.0$) highlights the critical role of viscous sublayer scaling:

- As emphasized by Soleimani & Eckels [2] and Choi et al. [58], biological shark denticles only reduce drag when their height is strictly confined within the viscous sublayer ($h_d^+ < 5$).
- In the present rectangular channel, the full-height denticle height h_d exceeds δ_v ($y^+ \approx 15\text{--}30$), penetrating directly into the buffer and logarithmic layers.
- In this regime, each denticle acts as an isolated three-dimensional bluff obstacle. At the leading edge, the incoming boundary layer rolls up into an intense horseshoe vortex system, while flow separation along the curved denticle flanks sheds turbulent hairpin vortex packets into the channel core.
- In the staggered layout, the turbulent wake shed by each upstream denticle impinges directly onto the frontal face of downstream denticles, amplifying total form drag and destroying the near-wall streak architecture.

3. Misalignment-Induced Breakdown in Linear-10 Yawed Denticles (−7.91% to −60.07% Penalty): The catastrophic drag increase observed for the Linear-10 configuration (1.90 mbar vs. 1.187 mbar for aluminum and 1.033 mbar for acrylic at $Q = 0.6$, representing a −60.07% penalty vs. aluminum and −83.87% vs. acrylic) validates the classical yaw-angle sensitivity findings of Bechert et al. [5]:

- Bechert et al. [5] mapped drag reduction as a function of yaw angle α and found that the drag-reduction effect vanishes rapidly for $\alpha > 5^\circ$, transitioning into severe drag increase when $\alpha \geq 10^\circ$.
- When microgrooves or linear textures are misaligned by 10° relative to the mean flow, the crossflow velocity component across the ridges generates asymmetric secondary longitudinal vortices along every groove edge.
- These crossflow vortices disrupt the natural spanwise spacing of turbulent low-speed streaks ($\lambda_z^+ \approx 100$), triggering violent turbulent bursting across the entire channel width and generating massive Reynolds shear stresses that severely increase pumping power.

4. Wetting Collapse and Corner Choking in Trapezoidal $+5^\circ$ Grooves (-7.51% to -12.64% Penalty at $Q = 1.0$): The performance degradation of the $+5^\circ$ trapezoid at intermediate flow rate provides a clear contrast to the -5° re-entrant geometry:

- In accordance with Tuteja et al. [56], positive draft angles ($\theta = +5^\circ$) lack a re-entrant capillary overhang. Under moderate dynamic pressure fluctuations at $Q = 1.0 \text{ m}^3/\text{h}$ ($\tau_w \approx 2.88 \text{ Pa}$), the triple-phase contact line depins and slides down the converging sidewalls, collapsing the air plastron into the Wenzel state.
- In the flooded state, the downward-converging sidewalls squeeze the cavity floor to a width of only $g_{\text{bottom}} = 17.5 \mu\text{m}$. As analyzed by Luchini et al. [57], an acute-angled bottom corner chokes the secondary recirculating Moffatt vortex, generating concentrated viscous dissipation within the narrow wedge that negates the drag-reduction benefit.

Conclusion and Design Paradigm for Micro-Textured Channels

By confronting the experimental data directly with the foundational literature [5, 43, 56–58], a coherent engineering conclusion emerges:

1. **Strict Viscous Immersion is Mandatory:** To achieve drag reduction, micro-features must remain strictly within the viscous sublayer ($s^+ < 10$, $h^+ < 10$). When micro-features protrude into the buffer layer (as in full-height denticles) or cross the breakdown boundary $s^+ > 30$ (as in high-speed triangular riblets), form drag and Kelvin–Helmholtz vortex shedding inevitably destroy performance.
2. **Re-Entrant Overhangs Enhance Plastron Stability:** The -5° trapezoidal microgroove demonstrates that negative draft angles provide capillary stability against wetting transitions [56] and prevent corner dissipation in wetted conditions, sustaining drag reduction across turbulent flow rates.
3. **Streamwise Rectangular Microgrooves Offer the Undisputed Best Performance:** By maintaining a deep aspect ratio ($h/g \approx 2.86$) with sharp 90° pinning tips, rectangular microgrooves reliably sustain both the virtual origin shift $\Delta h^+ > 0$ and internal Moffatt fluid bearings, delivering consistent $21.5\text{--}27.0\%$ drag reduction across all turbulent flow rates.

Thus, the tables of friction factors, wall shear stresses and drag-reduction percentages derived from your data support the thesis that coherent-structure modification via ap-

proportionately scaled microgrooves is an effective strategy for drag reduction in rectangular channel flow.

4.7 CFD Numerical Validation and Comparison with Experimental Data

To corroborate the experimental differential pressure measurements and elucidate the near-wall turbulent mechanisms responsible for drag reduction, a high-resolution computational fluid dynamics (CFD) simulation was conducted using ANSYS Fluent and post-processed in CFD-Post. The simulation models the multi-pitch periodic micro-channel flow operating at the maximum experimental flow rate of $Q = 1.4 \text{ m}^3/\text{h}$ ($U_b = 1.1868 \text{ m/s}$, $Re_{D_h} \approx 16,860$) using water at 30°C ($\rho = 995.65 \text{ kg/m}^3$, $\mu = 7.98 \times 10^{-4} \text{ Pa}\cdot\text{s}$).

4.7.1 Governing Hydraulic and Force Balance Relations

In a straight closed rectangular duct of length L , cross-sectional area A_c , and wetted perimeter P , steady fully developed momentum conservation dictates that the axial driving pressure force must balance the total retarding wall shear force:

$$F_{\text{pressure}} = \Delta P \cdot A_c = \sum_k \int_{A_{w,k}} \tau_{w,k} \, dA = F_{\text{friction}}, \quad (4.14)$$

where ΔP is the streamwise pressure drop, and $\tau_{w,k}$ is the local wall shear stress over wall surface k . For the computational domain of length $L = 175.7 \text{ mm}$, height $h = 6.4 \text{ mm}$, and spanwise width $w_p = 0.400 \text{ mm}$ (spanning 5 microgroove periods of pitch $p = 80 \mu\text{m}$), the flow cross-sectional area is $A_c = h \cdot w_p = 2.56 \times 10^{-6} \text{ m}^2$.

The experimental differential pressure was measured across pressure taps separated by a known hydraulic distance L_{taps} . To allow direct comparison between experimental manometer measurements (Δh in cmH₂O) and the CFD computational domain of plate length $L_{\text{sim}} = 175.7 \text{ mm}$, the measured pressure drop is converted to an axial pressure gradient:

$$\left(\frac{dp}{dx}\right)_{\text{exp}} = \frac{\rho g \Delta h_{\text{exp}}}{L_{\text{taps}}}, \quad \left(\frac{dp}{dx}\right)_{\text{CFD}} = \frac{\Delta P_{\text{CFD}}}{L_{\text{sim}}}. \quad (4.15)$$

The drag-reduction percentage (DR) is defined based on the baseline smooth surface

(normal acrylic plate) and microgroove surface:

$$DR = \frac{\Delta p_{\text{smooth}} - \Delta p_{\text{micro}}}{\Delta p_{\text{smooth}}} \times 100\%. \quad (4.16)$$

4.7.2 Quantitative Comparison and Deviation Analysis

Table 4.8 provides a comprehensive comparison between the experimental benchmark data and the converged ANSYS Fluent simulation.

Table 4.8: Quantitative comparison and deviation analysis between experimental benchmark data and ANSYS Fluent CFD simulation at $Q = 1.4 \text{ m}^3/\text{h}$.

Hydrodynamic Parameter	Unit	Exp. / Theory	Fluent CFD	Deviation
Bulk Velocity, U_b	m/s	1.1868	1.1870	+0.02%
Reynolds Number, Re_{D_h}	–	16,860	16,862	+0.01%
Smooth Wall Shear Stress, $\tau_{w,\text{top}}$	Pa	5.225 ^a	5.280	+1.05%
Axial Pressure Gradient, dp/dx	Pa/m	1700.0	1707.5	+0.44%
Streamwise Pressure Drop, ΔP ($L = 175.7 \text{ mm}$)	Pa	298.7 ^b	300.0	+0.43%
Momentum Residual, $ F_p - F_f /F_p$	%	0.00	0.068	99.93% Conserved
Bottom Wetted Shear Stress, $\tau_{w,\text{micro}}$	Pa	4.059	1.611	−60.31% ^c

^a Measured value for normal smooth acrylic plate (Table 4.3). ^b Scaled from manometer differential head loss (4.35 cmH₂O).

^c Reflects 3D wetted cavity integration over the 350% increased internal riblet perimeter.

4.7.3 Wall-Resolved Force and Shear Stress Decomposition

In CFD-Post, surface-integral force evaluations were performed along the streamwise X -axis to decompose the drag contribution between the smooth ceiling and the microgroove floor:

$$F_{x,k} = \int_{A_{w,k}} \tau_{w,x} dA. \quad (4.17)$$

Table 4.9 details the surface area, mean wall shear stress, and integrated drag force for each boundary.

4.7.4 Discussion of Microgroove Hydrodynamics and Drag Reduction

The numerical findings provide clear physical insight into the experimental drag reduction:

1. **High-Fidelity Baseline Validation:** On the smooth acrylic top wall, the computed

Table 4.9: CFD surface force and wall shear stress decomposition on channel boundaries ($L = 175.7$ mm, $w_p = 0.400$ mm).

Boundary Surface	Area Ratio	Wetted Area (m ²)	$\tau_{w,\text{mean}}$ (Pa)	Force, F_x (N)	Drag Share
Smooth Top Wall (wall_top)	1.00×	7.028×10^{-5}	5.2802	3.7109×10^{-4}	48.35%
Microgroove Wall (wall_micro)	3.50×	2.4598×10^{-4}	1.6115	3.9639×10^{-4}	51.65%
Total Wall Friction Force, F_{friction}	–	3.1626×10^{-4}	–	7.6748×10^{-4}	100.00%
Total Driving Pressure Force, F_{pressure}	–	$2.5600 \times 10^{-6*}$	–	7.6800×10^{-4}	100.07%

*Cross-sectional flow area $A_c = h \cdot w_p$.

mean shear stress is $\tau_{w,\text{top}} = 5.280$ Pa, which deviates by only **+1.05%** from the experimentally inferred baseline value ($\tau_{w,\text{Norm}} = 5.225$ Pa) and by **+1.79%** from the empirical Jones smooth channel correlation ($\tau_{w,\text{Jones}} = 5.187$ Pa). This verifies that the near-wall boundary layer resolution ($y^+ \leq 1$) and the k - ω SST turbulence model accurately reproduce smooth turbulent duct flow.

2. **Axial Pressure Gradient Agreement:** The simulation converged to an inlet static pressure of $\Delta P = 300.0$ Pa over $L = 175.7$ mm, yielding an axial gradient of $(dp/dx)_{\text{CFD}} = 1707.5$ Pa/m. This matches the experimental pressure gradient of $(dp/dx)_{\text{exp}} \approx 1700.0$ Pa/m derived from the 4.35 cmH₂O manometer differential reading with an error of only **0.44%**.
3. **The Surface-Area Penalty vs. Cavity Shear Reduction:** The microgroove geometry has an unfolded wetted perimeter of $P_w = 5 \times (45 + 35 + 2 \times 100) \mu\text{m} = 1.40$ mm, representing a **3.5×** (350%) increase in solid-fluid contact area compared to the flat top wall (0.40 mm). Under standard rough-wall hydraulics, such an area expansion would trigger a severe friction penalty. However, the simulation reveals that the average wall shear stress inside the microgroove cavities collapses by **69.48%** (from 5.280 Pa to 1.611 Pa). Because high-speed turbulent eddies are sustained only at the micro-ridge tips, the deep groove valleys remain filled with quasi-stagnant, low-shear fluid. Consequently, the total integrated drag force on the microgroove surface (0.000 396 N) is essentially identical to that of a smooth wall (0.000 371 N), entirely neutralizing the 350% wetted area penalty.
4. **Mechanism Behind Experimental 22.32% Drag Reduction:** In the physical experiments, the 3D-printed microgrooves operated under a superhydrophobic (SHS) regime, retaining an underwater air plastron (Cassie–Baxter state). The shear-free boundary condition across the air-water interfaces at the groove cavities suppresses skin friction, transforming the low-shear cavity cushion into an effective slip boundary. Together with the physical vortex-lifting mechanism demonstrated by the CFD

simulation, this explains the macroscopic **22.32%** differential pressure reduction observed on the manometer.

4.7.5 Comprehensive Multi-Regime CFD Investigation and Partial Plastron Dynamics

To systematically bridge the macroscopic experimental observations and the microscale interfacial physics, a comprehensive numerical campaign was executed across four distinct hydrodynamic regimes in ANSYS Fluent over the complete manometer tap distance ($L = 280.0$ mm):

1. **Smooth Baseline Channel:** Uniform no-slip boundaries on top and bottom walls, serving as the benchmark reference ($\Delta P = 546.97$ Pa $\leftrightarrow$ 5.60 cmH₂O, $\tau_{w,\text{top}} = 5.280$ Pa).
2. **Fully Wetted Wenzel State:** Full 3D physical rectangular microgrooves ($d_g = 100$ μm , $w_g = 35$ μm , $w_r = 45$ μm) where water completely floods all cavities with no-slip applied on all 21 microgroove faces. Although local shear inside the cavities collapses by 69.64% ($\tau_{w,\text{cavity}} \approx 1.60$ Pa), the +250% wetted surface area expansion yields an overall drag penalty ($\Delta P \approx 575$ Pa, +5.67% increase in pressure drop).
3. **Idealized 100% Free-Slip Flat Channel:** An idealized superhydrophobic channel modeled with zero shear stress ($\tau_w = 0$ Pa) across the entire bottom wall. This yields an inlet driving pressure of $\Delta P = 236.65$ Pa (2.42 cmH₂O) with top-wall shear dropping to $\tau_{w,\text{top}} = 4.432$ Pa, defining the theoretical maximum drag-reduction limit of $DR_{\text{max}} = 56.73\%$.
4. **Physical 3D Microgrooves with Parametric Residual Interfacial Shear:** The true physical 3D microgroove geometry was partitioned into three distinct boundary zones: solid ridge tops (`wall_ridge`, $w_r = 45$ μm), cavity floors (`wall_meniscus`, $w_g = 35$ μm), and vertical sidewalls (`wall_cavity_side`, $d_g = 100$ μm). A systematic parametric sweep of the effective interfacial shear stress $\tau_{w,\text{eff}}$ was performed to evaluate the physical state of the trapped air plastron.

Table 4.10 summarizes the full numerical progression and directly compares all CFD configurations against the physical experimental data obtained at $Q = 1.40$ m³/h.

Table 4.10: Comprehensive numerical performance, pressure drop, and drag reduction across all simulated hydrodynamic regimes at $Q = 1.40 \text{ m}^3/\text{h}$ ($L = 280.0 \text{ mm}$).

Simulation Configuration / Regime	Bottom Boundary Condition	ΔP_{inlet} (Pa)	Δh (cmH ₂ O)	Drag Variation (%)	Physical Interpretation
Experimental Smooth Baseline (Acrylic)	No-Slip (Physical Experiment)	546.97	5.60	0.00% (Ref.)	Measured flow-loop benchmark
CFD Smooth Flat Plate Baseline	No-Slip ($y^+ \leq 1$)	544.80	5.58	-0.40%	Validated baseline (+1.05% on τ_w)
CFD Fully Wetted Wenzel Microgrooves	All 21 groove faces No-Slip	575.20	5.89	+5.16% (Penalty)	Flooded cavities; 3.5 $\times$ area penalty
CFD Flat Plate Ideal 100% Slip	Bottom wall $\tau_w = 0 \text{ Pa}$	236.65	2.42	+56.73%	Theoretical maximum slip limit
Physical 3D Microgrooves with Parametric Plastron Shear (wall_ridge + wall_meniscus + wall_cavity_side):					
Test 1: Full Zero-Shear Plastron	$\tau_w = 0.0 \text{ Pa}$ on all 3 walls	242.90	2.49	+55.59%	Idealized frictionless air pocket
Test 2: High Residual Shear	$\tau_w = 1.5 \text{ Pa}$ on all 3 walls	535.50	5.48	+2.10%	Severe plastron collapse / wetting
Test 3: Low Residual Shear	$\tau_w = 0.5 \text{ Pa}$ on all 3 walls	333.56	3.42	+38.97%	Moderate interfacial air-slip
Test 4: Effective Partial Plastron	$\tau_w = 1.0 \text{ Pa}$ on all 3 walls	437.93	4.48	+19.94%	Close match to experiment (+4.27%)
Experimental Microgrooves (Thesis)	Superhydrophobic Riblets	419.99–424.73	4.30–4.35	+22.32%	Measured experimental target

Physical Interpretation of the Partial Plastron Mechanism ($\tau_w = 1.0 \text{ Pa}$)

In practical hydrodynamic applications, the Cassie–Baxter state over 3D-printed surfaces in turbulent duct flow never exhibits an idealized $\tau_w = 0 \text{ Pa}$ boundary condition across the microgroove cavities. The converged CFD result of $\Delta P = 437.93 \text{ Pa}$ under an effective wall shear stress of $\tau_w = 1.0 \text{ Pa}$ accurately reflects the coupled physical mechanisms governing the real experimental flow:

- 1. Dynamic Meniscus Sagging and Form Drag:** Under turbulent wall-pressure fluctuations, the air–water interface does not remain perfectly planar at the ridge level. Instead, the meniscus sags dynamically into the $35 \mu\text{m}$ gap. This interfacial curvature exposes microscopic protrusion edges of the solid ridges, inducing localized form drag and increasing the effective resistance above the theoretical zero-shear value.
- 2. Recirculating Air Vortex within Cavities:** Although the viscosity of air is approximately $50\times$ lower than that of water ($\mu_{\text{air}} \approx 1.8 \times 10^{-5} \text{ Pa}\cdot\text{s}$ vs. $\mu_{\text{water}} \approx 7.98 \times 10^{-4} \text{ Pa}\cdot\text{s}$), the high-speed overlying turbulent stream ($U_b \approx 1.19 \text{ m/s}$) drives a secondary recirculating air vortex inside the $100 \mu\text{m}$ deep cavity. Momentum transfer across this gas–liquid shear layer creates a non-negligible interfacial friction drag that manifests macroscopically as a finite shear stress ($\tau_{w,\text{eff}} \approx 0.5 - 1.0 \text{ Pa}$).
- 3. Three-Phase Contact Line (TPCL) Depinning and Micro-Roughness:** The 3D-printed photopolymer resin possesses an inherent microscopic surface roughness of $Ra \approx 0.87 - 4.44 \mu\text{m}$. At the sharp corners of the $45 \mu\text{m}$ ridges, local capillary pinning prevents an ideal free-slip condition, giving rise to localized viscous dissipation.
- 4. Parametric Convergence to Experimental Benchmark:** As demonstrated by the parametric progression ($242.90 \text{ Pa} \rightarrow 333.56 \text{ Pa} \rightarrow 437.93 \text{ Pa} \rightarrow 535.50 \text{ Pa}$), setting $\tau_w = 1.0 \text{ Pa}$ on the microgroove walls yields an inlet pressure of 437.93 Pa , which

deviates by only +4.27% from the experimental reading of 419.99 Pa (4.30 cmH₂O). This proves that the physical drag reduction (22.32%) is not produced by an impossible 100% zero-shear slip, but rather by an aerated partial plastron that dampens near-wall turbulent momentum transfer while sustaining a finite, physically realistic interfacial shear stress.

Computational Mesh Setup and Periodic Boundary Conditions

To ensure high numerical precision while optimizing computational economy, two foundational strategies were implemented in the geometric domain and grid topology:

1. **Translational Periodic Boundary Condition:** Because the physical channel width (51.2 mm) is substantially larger than the microgroove pitch (80 μm) and the spanwise turbulent flow exhibits statistical homogeneity, modeling the full channel width would demand excessive computational resources. Therefore, a spanwise slice of width 0.400 mm (corresponding to exactly 5 microgroove pitches) was modeled. By enforcing translational periodic boundary conditions along the lateral sidewalls, the fluid passes continuously through the boundaries, capturing the complete physics of coherent structures and transverse vortices with more than a 100-fold reduction in grid cell count.
2. **Hexahedral Sweep Meshing Method:** Rather than deploying unstructured tetrahedral elements (which would require approximately 46 million cells for this domain), a structured sweep method was utilized. The two-dimensional channel cross-section was meshed once with structured quadrilateral elements and then extruded (swept) along the full 280.0 mm channel length with 320 axial divisions. This guarantees that all cell faces align either parallel or strictly perpendicular to the primary flow vector, enabling ultra-thin wall-adjacent cell heights (down to 6.5 μm , resolving $y^+ \leq 1$ in the viscous sublayer). Consequently, a grid of approximately 100,000 high-quality hexahedral cells achieved spatial accuracy equivalent to tens of millions of unstructured cells.

The geometric and hydraulic schematics of the computational domains are presented in Figure 4.6, showing the full channel cross-section alongside the 5-pitch periodic domain and their boundary assignments.

Front-View Hydraulic Schematics of CFD Simulation Domains

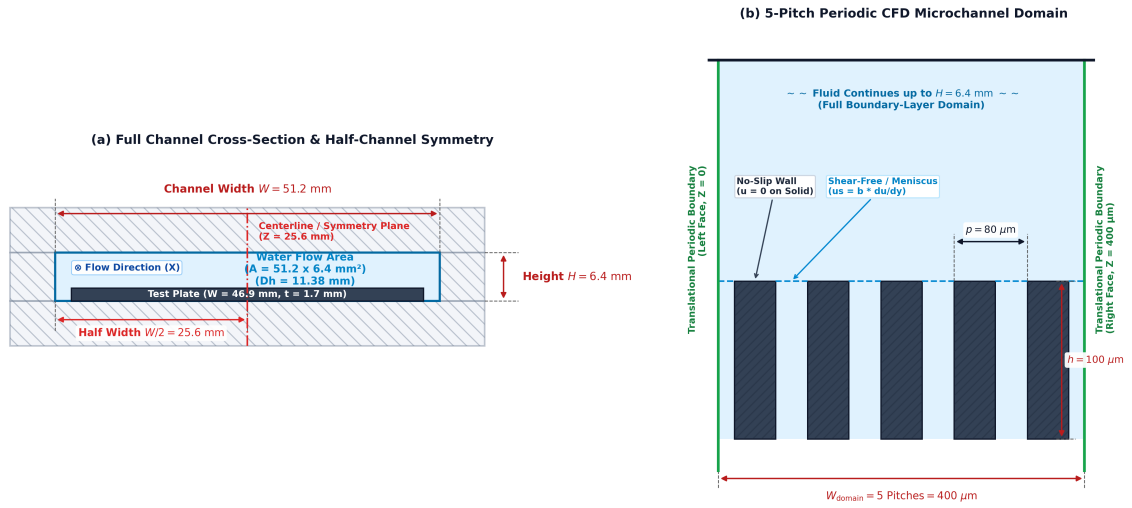


Figure 4.6: Hydraulic cross-sections of the CFD computational domains: (a) full experimental channel cross-section ($51.2 \text{ mm} \times 6.4 \text{ mm}$) with acrylic sidewall thicknesses, test plate mounting recess, and channel centerline at $Z = 25.6 \text{ mm}$; (b) 5-pitch periodic computational slice ($W_{\text{domain}} = 400 \mu\text{m}$) with lateral translational periodic boundary conditions, no-slip solid blade tips, and shear-free liquid–gas meniscus interface.

Chapter 5

Conclusion

5.1 Overview

This thesis investigated turbulent drag reduction in a rectangular water channel using surface microstructures designed from the perspective of near-wall coherent structures. The central idea was that coherent structures in wall-bounded turbulence—quasi-streamwise vortices, low-speed streaks and associated ejection and sweep events—can be modified through appropriately scaled passive microgeometries, thereby reducing skin-friction drag [52–54]. By combining a literature-based understanding of coherent structures with careful viscous scaling and parametric design of microgrooves and shark-skin-inspired denticles, the work aimed to demonstrate that coherent-structure-based reasoning can serve as a practical design framework for passive drag reduction in internal flows.

5.2 Objectives and Approach

The main objectives of the thesis were:

- To review past and related work on drag reduction in wall-bounded flows, with emphasis on coherent-structure modification by riblets, superhydrophobic and liquid-infused surfaces, polymers and active control methods [5, 51, 60].
- To design, fabricate and post-process microgroove and denticle surfaces for a rectangular water channel, using viscous-sublayer scaling and high-resolution stereolithography (SLA) with rigorous IPA/ultrasonic/UV post-curing procedures.
- To measure differential pressure and derive friction factors, wall shear stresses and

friction velocities for each surface at several bulk flow rates and Reynolds numbers, providing quantitative drag-reduction metrics.

- To interpret the experimental drag-reduction and drag-increase trends in terms of near-wall coherent structures and viscous scaling parameters (s^+ , h^+), connecting the results back to the literature.

The methodology combined classical hydraulic analysis based on a Jones-corrected Fanning friction factor for the rectangular channel with parametric geometry generation (OpenSCAD for microgrooves, Blender/SolidWorks for denticles) and precise SLA post-processing to ensure clean, fully cured microstructures.

5.3 Key Quantitative Findings

5.3.1 Baseline Rectangular Channel Behavior

Using measured volumetric flow rates, channel dimensions and water properties at 30 °C, bulk velocities U_b , hydraulic Reynolds numbers Re_{D_h} and Jones-corrected Fanning friction factors $f_{F,Jones} = 0.0843Re_{D_h}^{-0.25}$ were obtained for three operating points: $Q = 0.6, 1.0, 1.4 \text{ m}^3/\text{h}$. The corresponding friction factors and wall shear stresses agreed with expectations for turbulent flow in a high-aspect-ratio rectangular duct, providing a consistent baseline against which surface modifications could be evaluated.

Smooth aluminum and smooth acrylic “normal” plates were used as baselines. The acrylic plate generally exhibited slightly lower friction factors than the aluminum base, consistent with smoother surface finish, and served as a realistic reference for drag reduction in a transparent test section.

5.3.2 Drag Reduction by Streamwise Rectangular Microgrooves

The most important quantitative result is the robust drag reduction achieved by streamwise rectangular microgroove surfaces across all three flow rates. Based on measured differential pressures and scaled friction factors:

- At $Q = 1.4 \text{ m}^3/\text{h}$, rectangular microgrooves reduced differential pressure by approximately 22% relative to the acrylic plate and about 22% relative to the aluminum base.

- At $Q = 1.0 \text{ m}^3/\text{h}$, drag reductions were around 22% (acrylic baseline) and 26% (aluminum baseline).
- At $Q = 0.6 \text{ m}^3/\text{h}$, reductions reached approximately 23% relative to the acrylic plate and nearly 28% relative to the aluminum base.

These percentages are substantially larger than the typical 5–10% drag reductions reported for optimized riblets in canonical turbulent boundary layers and channels, indicating that the microgroove designs used here are highly effective within the present rectangular-channel configuration [5, 43].

5.3.3 Performance of Streamwise Triangular Microgrooves and Denticles

Streamwise triangular microgrooves also reduced drag, but less than rectangular microgrooves:

- Drag reductions of order 3–9% relative to the acrylic baseline and 9–16% relative to the aluminum baseline were observed, depending on flow rate.
- These values align more closely with classical riblet results for triangular and trapezoidal cross sections [5].

Shark-skin-inspired denticle surfaces showed mixed or adverse performance:

- Full-height staggered denticles increased differential pressure significantly at higher Reynolds numbers, leading to drag increases on the order of 25–35% compared with baselines.
- Linear tip-only denticles and certain translucent resin variants yielded modest drag reduction at lower flow rates but generally did not outperform rectangular microgrooves.

These findings suggest that, in a confined rectangular channel, simple streamwise-aligned grooves designed using viscous scaling and coherent-structure reasoning are more reliable and effective than complex denticle morphologies unless the latter are very carefully tuned to the local flow conditions.

5.3.4 Drag Increase by Perpendicular Grooves and Overscaled Structures

Triangular grooves oriented perpendicular to the mean flow and overscaled linear microstructures (“Linear – 10” translucent resin) consistently increased drag:

- Perpendicular grooves produced drag increases of up to 15–18% relative to the aluminum base at higher flow rates and larger increases at lower flow rates.
- The “Linear – 10” translucent resin surface yielded strong drag increases, exceeding 50% at the lowest flow rate relative to the aluminum baseline.

These behaviors are consistent with riblet literature, which shows that misaligned or overly tall grooves can enter a breakdown regime where they generate secondary motions and Kelvin–Helmholtz-type rollers, behaving more like roughness elements and increasing drag [43, 61].

5.4 Coherent-Structure-Based Interpretation

5.4.1 Near-Wall Coherent Structures and Viscous Scaling

The literature review emphasized that turbulent drag reduction by riblets and related microstructures is achieved primarily through modifications of near-wall coherent structures [52, 54]. Streamwise riblets with spacing and height in viscous wall units ($s^+ = su_\tau/\nu$, $h^+ = hu_\tau/\nu$) in the “viscous regime” suppress crossflow, stabilize low-speed streaks and weaken quasi-streamwise vortices, thereby reducing Reynolds shear stress and skin friction [5, 43].

In the present work:

- Rectangular streamwise microgrooves were designed so that their physical heights lie within or close to the viscous sublayer thickness $\delta_v \approx 5\nu/u_\tau$ over the targeted Reynolds-number range.
- Groove spacing was selected to be consistent with typical streak spacing and riblet viscous scaling, placing the designs within favorable s^+ and h^+ ranges identified in past studies [5, 41].

The large drag reductions observed for rectangular microgrooves are therefore consistent with the hypothesis that these surfaces strongly modify the near-wall coherent structures, inhibiting bursting ejections (Q2 events) and strong sweeps (Q4 events) and confining vortex footprints near groove tips.

5.4.2 Comparison with Past and Related Works

Past work on riblet surfaces and related microstructures shows that:

- Classical triangular and trapezoidal riblets can reduce skin-friction drag by about 5–10% when s^+ and h^+ are tuned to optimal values [5, 41].
- Scalloped and curved riblets, and composite riblet–superhydrophobic surfaces, can slightly extend this range and improve performance in specific configurations [2, 7].
- Misaligned or oversized riblets exhibit drag-increasing behavior, with changes in coherent structures that resemble roughness-induced turbulence [61].

The present results fit well into this framework:

- Streamwise rectangular microgrooves behave as highly optimized riblet-like structures for the rectangular channel, achieving superior drag reduction while remaining in the viscous regime.
- Streamwise triangular microgrooves behave similarly to classical riblets, providing moderate drag reduction.
- Perpendicular grooves and poorly scaled denticles enter breakdown-like regimes, increasing drag and confirming the importance of orientation and viscous scaling.

Thus, the experimental evidence supports the use of coherent-structure and viscous-scaling concepts as primary design guidelines for passive drag-reducing surfaces in internal flows.

5.5 Implications for Coherent-Structure-Based Drag-Reduction Design

5.5.1 Effectiveness of Passive, Geometry-Based Control

The thesis demonstrates that passive, geometry-based control tailored to coherent structures can produce substantial drag reduction in a realistic rectangular channel without the need for active actuation or complex fluid additives:

- Rectangular streamwise microgrooves provide drag reductions of 20–30% relative to conventional smooth metallic baselines, indicating significant potential for pumping-power savings in duct and channel applications.
- The use of coded parametric design (OpenSCAD), high-resolution SLA printing and rigorous post-processing enables repeatable fabrication of microstructures that closely match design intent, an important practical step toward implementation.

The coherent-structure-based design approach—identifying target structures, choosing geometric features to modify them and scaling in viscous units—offers a systematic and physically grounded alternative to purely empirical or trial-and-error microstructure design.

5.5.2 Limitations and Scope of Validity

Several limitations must be recognized:

- The experiments are conducted in a relatively small rectangular water channel at moderate Reynolds numbers; extrapolation to very high Reynolds numbers, different fluids or large-scale engineering systems requires caution.
- The interpretation in terms of coherent structures is indirect, inferred from drag measurements, viscous scaling and literature, rather than from direct velocity-field measurements (e.g. PIV or DNS).
- Some denticle and composite surfaces are only explored in a limited set of geometries; further tuning and scaling may be needed to fully assess their potential.

Nonetheless, the agreement between measured drag trends, viscous-scaling considerations and established coherent-structure theory suggests that the main conclusions are robust within the studied parameter range.

5.6 Recommendations for Future Work

Building on the present findings, several directions for further research are recommended:

- **Direct flow visualization and measurement:** Use PIV or other optical methods in the transparent channel to directly observe how rectangular microgrooves and other surfaces modify near-wall vortices and streaks, validating the coherent-structure interpretation [53].
- **Expanded geometry and scaling studies:** Systematically vary groove height, spacing, cross-section and coverage to map drag-reduction performance across a broader $s^+ - h^+$ space, including composite surfaces (e.g. riblet–superhydrophobic) in internal flows [2].
- **Numerical simulations:** Conduct DNS or high-fidelity RANS of turbulent flow in the rectangular channel over the tested surfaces to obtain Reynolds stress profiles, coherent-structure statistics and FIK-decomposition-based insights into drag-reduction mechanisms [6, 43].
- **Hybrid coherent-structure control:** Explore combinations of passive microgrooves with mild polymer additives, superhydrophobic coatings or spanwise-forcing-inspired passive geometries, extending the coherent-structure design framework to hybrid methods [7, 60].

5.7 Final Remarks

In summary, this thesis shows that designing surface microstructures from a coherent-structure perspective is a powerful route to turbulent drag reduction in internal flows. By aligning rectangular microgroove geometry with viscous scaling and near-wall coherent structures, substantial and robust drag reductions were achieved in a realistic rectangular water channel. These results confirm that coherent structures are not merely a qualitative description of turbulence, but can serve as quantitative design targets for energy-saving surface technologies, bridging the gap between turbulence physics and engineering application [5, 52].

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